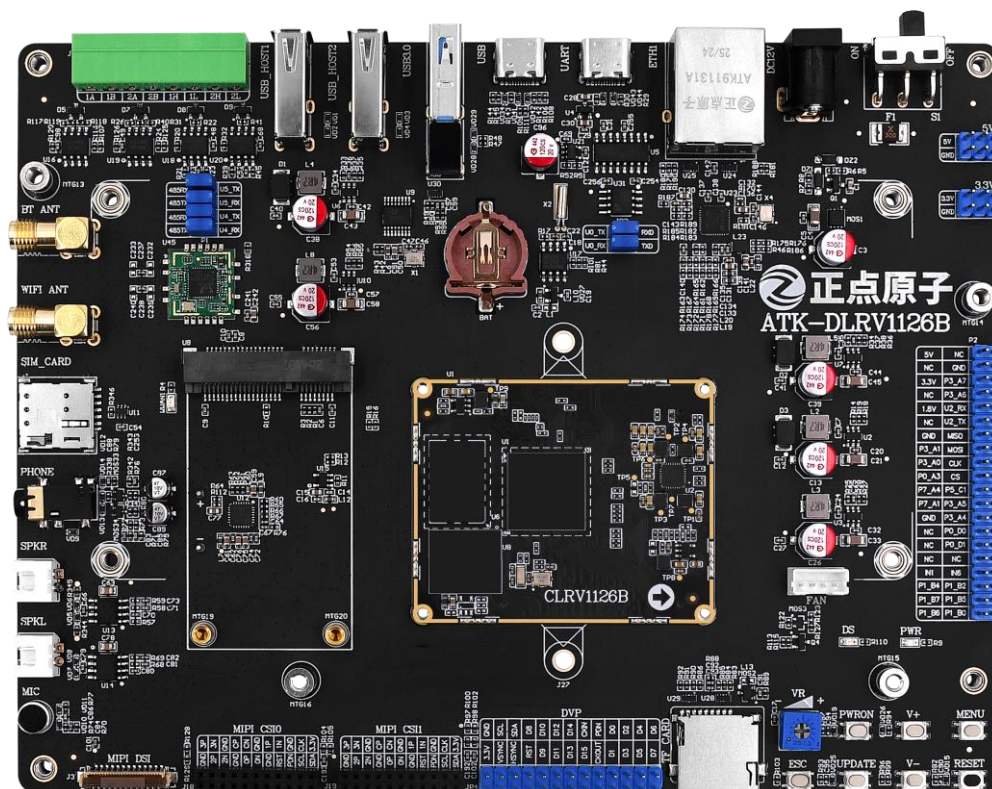


ATK-DLRV1126B

Buildroot Quick test manual

V1.0



1. Shopping:TMALL: <https://zhengdianyuanzi.tmall.com>TAOBAO: <https://openedv.taobao.com>**2. Download**Address: <http://www.openedv.com/docs/index.html>**3. FAE**Website : www.alientek.comForum : <http://www.openedv.com/forum.php>Videos : www.yuanzige.com

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Chapter 1. Preparation for using the ATK-DLRV1126B development board

Here we will introduce the matters that need to be paid attention to when using the development board. Please read carefully. Because if you don't pay attention to these matters, it may lead to the failure of your experiment or damage to the development board, etc. So, please be sure to pay attention.

1.1 Precautions before powering on

Do not place the development board on a cluttered workbench. A protective film must be placed under the baseboard; otherwise, it may easily come into contact with metal foreign objects, causing a short circuit between two contacts on the bottom of the board.

After prolonged use, also check regularly for debris that may have fallen onto the development board to prevent short circuits. Keep the board away from water, moisture, and dust.

1.2 Serial Port Software Installation

Here is a brief introduction to the installation of the CH340 USB serial port driver and the MobaXterm terminal. Detailed installation tutorials are not provided here, as these are basic operations.

Install the MobaXterm terminal software (or other terminal software such as Xshell, SecureCRT, etc.). This document takes MobaXterm as an example. Double-click to open the archive, wait for extraction, then directly double-click MobaXterm_installer_v24.1.msi to install. The MobaXterm installer will guide you through the setup.

Install the CH340 USB serial port driver (this driver is required for the PC to communicate with the development board via the serial port).

Double-click to run it, click Install directly in the pop-up window, and wait until the pre-installation or successful installation window appears.

Note: The default serial port baud rate of the development board is 1500000 (1.5 M). Enter 1500000 directly in MobaXterm.

Then double-click to launch MobaXterm, and create a serial port session in the following order:

At step 3, select Serial from the drop-down menu.

At step 4, double-click and manually enter the baud rate 1500000 (**this baud rate is not preset in MobaXterm and must be entered manually**).

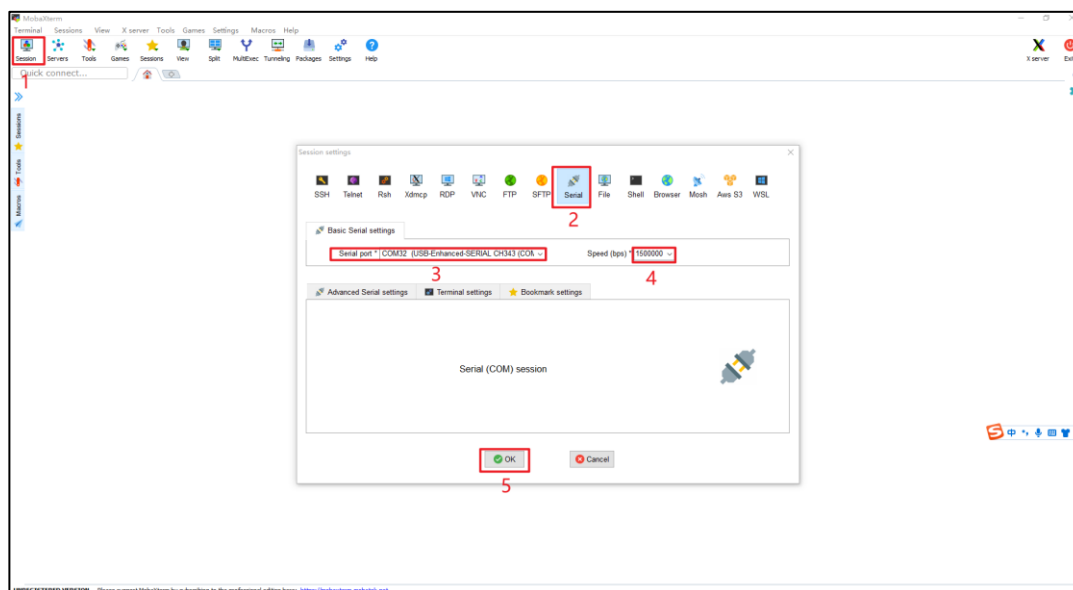


Figure 1.2-1 MobaXterm Serial Port Baud Rate Setting

1.3 Connecting the Development Board to the PC

During development, we usually interact with the development board through the serial port. Network connection is also available, but you need to know the IP address of the development board first. Therefore, the serial port is very important at this stage.

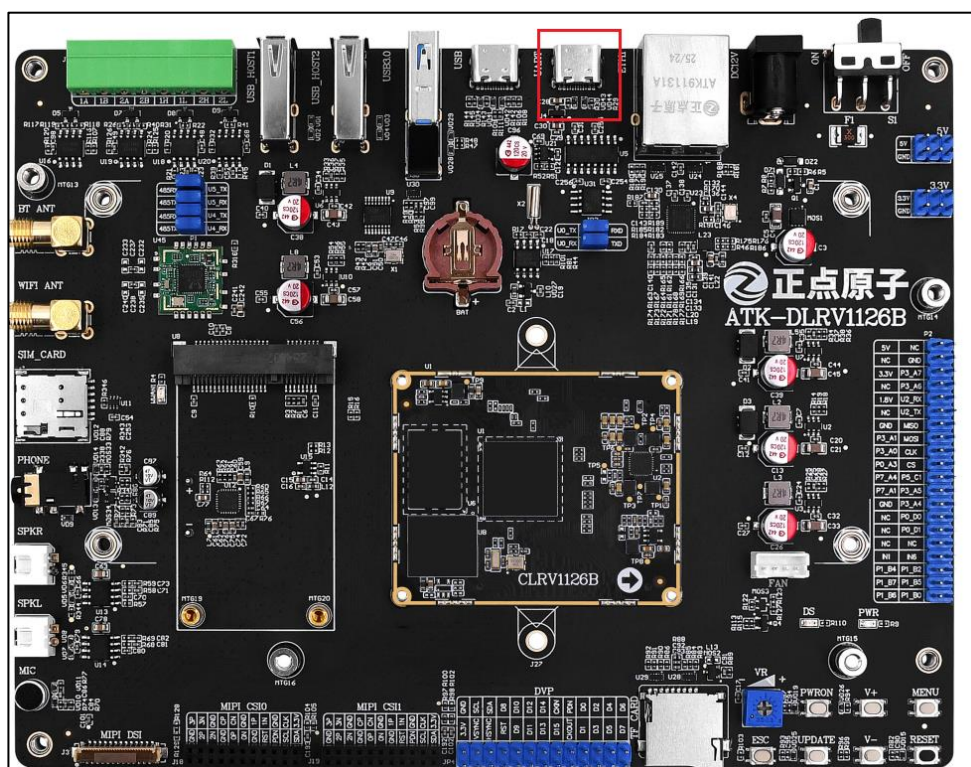


Figure 1.3-1 ATK-DLRV1126B Serial Port

The serial port output information is as follows (partial).

```

U-Boot 2017.09-g2b3ee34 #lixinkai (Nov 26 2025 - 09:51:40 +0800)

Model: ALIENTEK RV1126B board
MPIDR: 0x0
PreSerial: 0, raw, 0x20810000
DRAM: 2 GiB
System: init
Relocation Offset: 7d95e000
Relocation fdt: bb7faf10 - bb7fecdb
CR: M/C/I
Using default environment

DM: v2
mmc@21470000: 0
Best phase range 237-214 (31 len)
Successfully tuned phase to 45, used 1ms
Bootdev(atags): mmc 0
MMC0: HS200, 200Mhz
PartType: EFI
boot mode: None
RESC: 'boot', blk@0x0001b19f
resource: sha256+
FIT: no signed, no conf required
- dev=saradc, channel=5, dtb_adc=5545, read=2819, found=0
- dev=saradc, channel=5, dtb_adc=2760, read=2819, found=1
DTB: rv1126b-alientek-mip1080x1920#_saradc_ch5=2760.dtb
HASH(ce): OK
I2C0 speed: 100000Hz
PMIC: RK8013 (on=0x20, off=0x00, req_dvs: 0, act: high)
vdd_npu 950000 uV
vdd_logic 900000 uV
vdd_cpu init 950000 uV
use 0xbde00000 as drm logo base memory
Model: Alientek RV1126B Board
CLK: (uboot. arm: enter 0 KHz, init 0 KHz, kernel 0N/A)
gpll 1188000 KHz
aupll 786431 KHz
cpll 1000000 KHz
aclk_peri_root 200000 KHz
pclk_peri_root 100000 KHz
aclk_top_root 600000 KHz
pclk_top_root 100000 KHz
aclk_bus_root 400000 KHz
hclk_bus_root 200000 KHz
pclk_bus_root 100000 KHz
busclk_pmu_src 100000 KHz
Net: eth0: ethernet@21c70000
Hit key to stop autoboot('CTRL+C'): 0
## Booting FIT Image at 0xb8fc6100 with size 0x02633e00
Fdt Ramdisk skip relocation
## Loading kernel from FIT Image at b8fc6100 ...
Using 'conf' configuration
## Verified-boot: 0
Trying 'kernel' kernel subimage
Description: unavailable
Type: Kernel Image
Compression: uncompressed
Data Start: 0xb8ff3d00

```

Figure 1.3-2 ATK-DLRV1126B Boot Information

1.4 SSH login password

The default SSH login user for the ATK-DLRV1126B is root, and the password is also root.

By default, the serial port logs in automatically without entering a password upon startup; the root user logs in automatically.

Okay, you can now proceed to Chapter 3 for hardware testing.

Chapter 2. Development Board Function Testing

This chapter describes how to test the peripheral functions of the development board. It explains the usage and testing methods of the board, which is very helpful for development.

Note that many tests require additional test tools; please complete the test steps according to your actual situation. All hardware has been fully tested before delivery, so no further testing is required by the user.

2.1 LED Testing

The ATK-DLRV1126B is equipped with one user LED (DS). In the factory-default system, you can control the state of the LED using the following commands. After entering the commands, observe the state of the LED on the development board.

```
cat /sys/class/leds/work/trigger      # View the current trigger mode and supported
trigger modes of LED0; the default mode is [heartbeat].

echo none > /sys/class/leds/work/trigger  # Change the trigger mode of LED0.
echo 1 > /sys/class/leds/work/brightness  # Turn on LED0 (it will actually be off at this time).
Use a positive integer greater than 0; normally use 1.
echo 0 > /sys/class/leds/work/brightness  # Turn off LED0 (it will be in steady-on state at this
time). Set the value to 0.
```

2.2 Key Test

The ATK-DLRV1126B is equipped with 7 keys. Among them, the four keys V+, MENU, ESC, and V- are connected to channel 0 of ADC0. When a key is pressed, ADC0 can detect different voltage changes on channel 0.

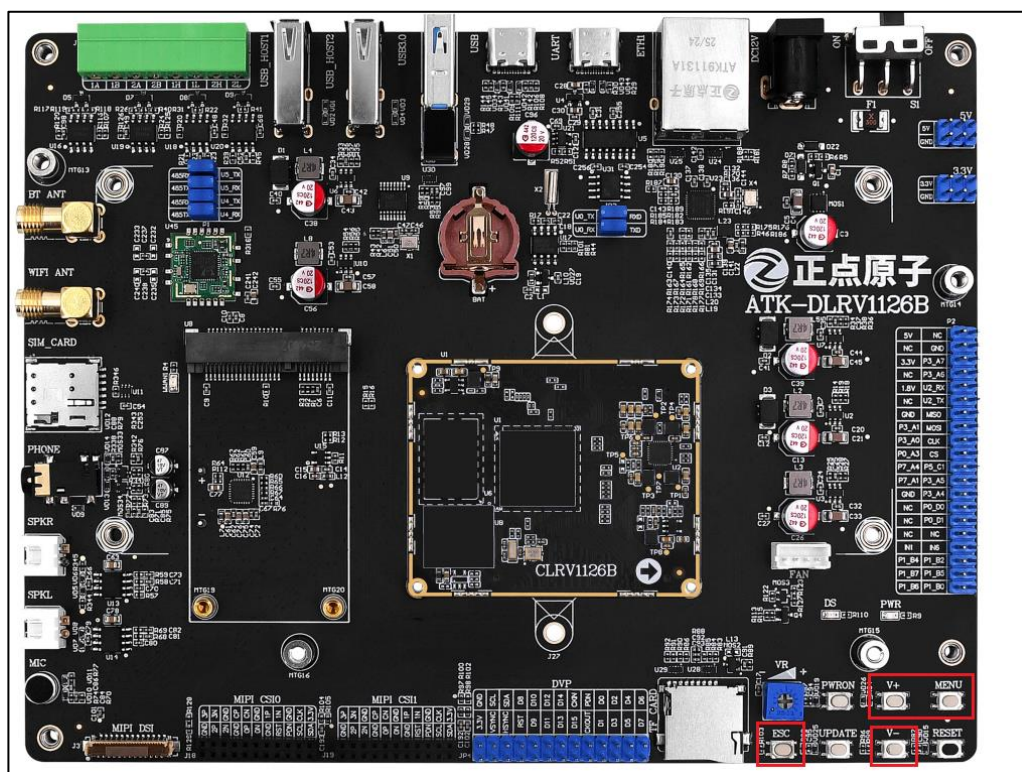


Figure 2.2-1 ATK-DLRV1126B Keys

Run the following command in the serial terminal of the development board to perform the test, then enter the number "4", because the key event is event4 (the event name shall prevail here; for example, the event name for key input is adc-keys, and the sequence number may change). To stop the test, press **Ctrl + C**.

Please press the above four keys on the development board. Do not press other keys such as the RESET key.

evtest

The running results are shown below. You can see that pressing the corresponding key will print the corresponding key type, with a value of 1 when pressed and 0 when released.

```

root@ATK-DLRV1126B:/# evtest
No device specified, trying to scan all of /dev/input/event*
Available devices:
/dev/input/event0:      rk805 pwrkey
/dev/input/event1:      ts_gt9xx
/dev/input/event2:      headset-keys
/dev/input/event3:      rockchip-es8390 Headset
/dev/input/event4:      adc-keys
Select the device event number [0-4]: 4
Input driver version is 1.0.1
Input device ID: bus 0x19 vendor 0x1 product 0x1 version 0x100
Input device name: "adc-keys"
Supported events:
Event type 0 (EV_SYN)
Event type 1 (EV_KEY)
Event code 1 (KEY_ESC)
Event code 114 (KEY_VOLUMEDOWN)
Event code 115 (KEY_VOLUMEUP)
Event code 139 (KEY_MENU)
Properties:
Testing ... (interrupt to exit)
Event: time 1609651608.026176, type 1 (EV_KEY), code 115 (KEY_VOLUMEUP), value 1
Event: time 1609651608.026176, ----- SYN_REPORT -----
Event: time 1609651608.130172, type 1 (EV_KEY), code 115 (KEY_VOLUMEUP), value 0
Event: time 1609651608.130172, ----- SYN_REPORT -----
Event: time 1609651608.754178, type 1 (EV_KEY), code 114 (KEY_VOLUMEDOWN), value 1
Event: time 1609651608.754178, ----- SYN_REPORT -----
Event: time 1609651608.962168, type 1 (EV_KEY), code 114 (KEY_VOLUMEDOWN), value 0
Event: time 1609651608.962168, ----- SYN_REPORT -----
Event: time 1609651610.522189, type 1 (EV_KEY), code 1 (KEY_ESC), value 1
Event: time 1609651610.522189, ----- SYN_REPORT -----
Event: time 1609651610.730180, type 1 (EV_KEY), code 1 (KEY_ESC), value 0
Event: time 1609651610.730180, ----- SYN_REPORT -----
Event: time 1609651612.290177, type 1 (EV_KEY), code 139 (KEY_MENU), value 1
Event: time 1609651612.290177, ----- SYN_REPORT -----
Event: time 1609651612.394168, type 1 (EV_KEY), code 139 (KEY_MENU), value 0
Event: time 1609651612.394168, ----- SYN_REPORT -----

```

Figure 2.2-2 Key Print Key Value

2.3 MIPI Screen Test

The factory system of the ATK-DLRV1126B development board supports three official MIPI screens from Punctual Atom store, including the 5.5-inch 720P MIPI screen, 5.5-inch 1080P MIPI screen, and 10.1-inch MIPI screen. The following shows the display effect of the 5.5-inch 720P MIPI screen; the actual effect is clearer.

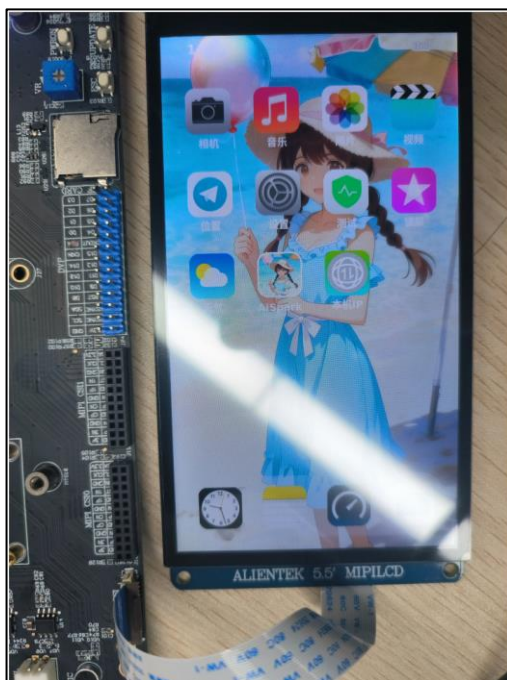


Figure 2.3-1 720P MIPI screen display effect
(the actual display content is subject to the factory system image).

2.4 Touch Test

After the development board boots up, we use the `evtest` command to check the corresponding touch event of the touch screen. The same method as in Section 3.2 can also be used to test whether the screen can touch normally. Enter the number **1**, and the corresponding driver for the screen is `ts_gt9xx`. Of course, you can also click the Qt interface in the factory system; normal touch operation also indicates that the touch function is working properly.

2.4.1 Multi-Touch Test

The MIPI screen supports up to 5-point touch. Although Qt supports multi-touch, the controls we use are single-touch. A dedicated multi-touch program is required to test multi-touch. Therefore, you cannot experience multi-touch on the factory default interface, so we still use the `evtest` command for testing.

Similarly, the same method as in Section 3.2 can be used to test whether the screen can touch normally. Enter the number 1, and the corresponding driver for the screen is `ts_gt9xx`.

```
evtest
```

```

root@ATK-DLRV1126B:~# evtest
No device specified, trying to scan all of /dev/input/event*
Available devices:
/dev/input/event0: rk805_nvrkey
/dev/input/event1: ts_gt9xx
/dev/input/event2: headset-keys
/dev/input/event3: rockchip-es8390 Headset
/dev/input/event4: adc-keys
Select the device event number [0-4]: 1
Input driver version is 1.0.1
Input device ID: bus 0x18 vendor 0x416 product 0x38f version 0x1060
Input device name: "ts_gt9xx"
Supported events:
Event type 0 (EV_SYN)
Event type 1 (EV_KEY)
Event code 59 (KEY_F1)
Event code 60 (KEY_F2)
Event code 61 (KEY_F3)
Event code 62 (KEY_F4)
Event code 63 (KEY_F5)
Event code 64 (KEY_F6)
Event code 125 (KEY_LEFTMETA)
Event code 330 (BTN_TOUCH)
Event type 3 (EV_ABS)
Event code 0 (ABS_X)
Value 0
Min 0
Max 1079
Event code 1 (ABS_Y)
Value 0
Min 0
Max 1919
Event code 47 (ABS_MT_SLOT)
Value 0
Min 0
Max 4
Event code 48 (ABS_MT_TOUCH_MAJOR)
Value 0
Min 0
Max 255
Event code 50 (ABS_MT_WIDTH_MAJOR)
Value 0
Min 0
Max 255
Event code 53 (ABS_MT_POSITION_X)
Value 0
Min 0
Max 1079
Event code 54 (ABS_MT_POSITION_Y)
Value 0
Min 0
Max 1919
Event code 57 (ABS_MT_TRACKING_ID)
Value 0
Min 0
Max 65535

```

Figure 2.4-1 Touch Test

2.4.2 MIPI Backlight Test

The backlight of the MIPI screen supports 255-level PWM adjustment. The brightness range is 0 to 255, with the default value being 255. The higher the value, the brighter the screen.

```
cat /sys/class/backlight/backlight/max_brightness
```

Check the maximum brightness level of the LCD.

```
cat /sys/class/backlight/backlight/brightness # Check the current brightness level.
```

echo 250 > /sys/class/backlight/backlight/brightness # Modify the current brightness level and observe the change in screen brightness.

```
cat /sys/class/backlight/backlight/brightness # Check the current brightness level again.
```

2.5 Audio Test

The ATK-DLRV1126B is equipped with the CODEC ES8390 audio chip. The ES8390 is a high-performance, low-power, low-cost stereo audio codec consisting of 2-channel ADC, 2-channel DAC, microphone amplifier, headphone amplifier, digital audio effects, as well as analog mixing and gain functions. The chip uses high-order multi-bit delta-sigma modulation technology for data conversion between digital and analog domains, resulting in low clock jitter sensitivity and low out-of-band noise.

The ATK-DLRV1126B development board has an onboard 3.5mm headphone jack, an onboard microphone, and an onboard speaker. Hot-plug switching is supported among the headphone, microphone, and speaker. We use these three components for audio testing.

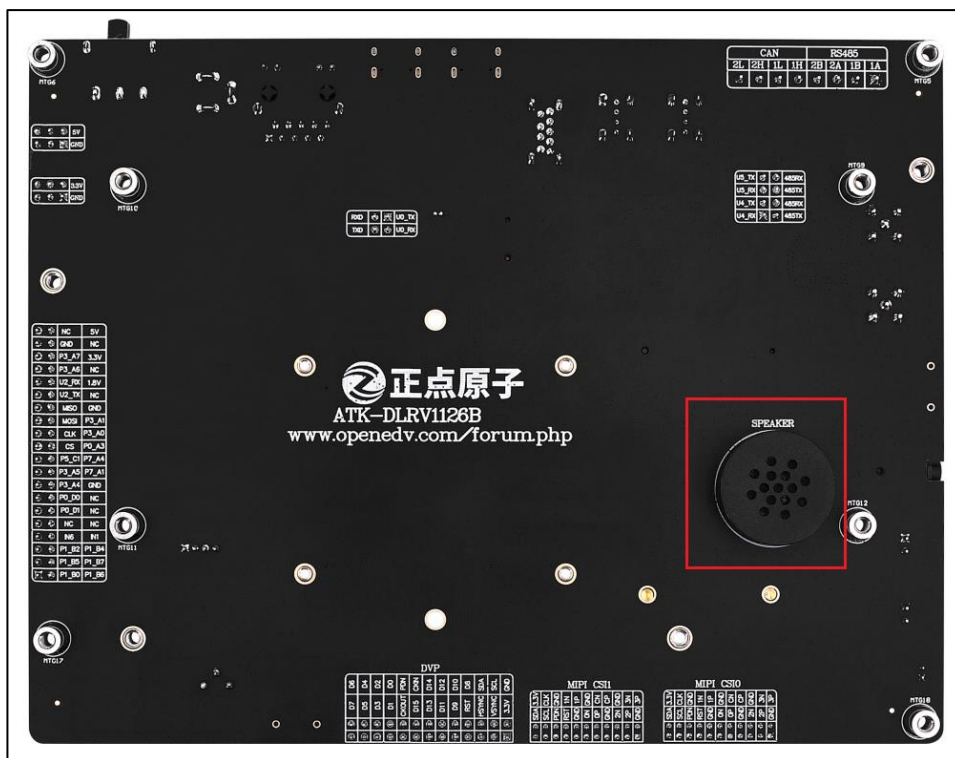
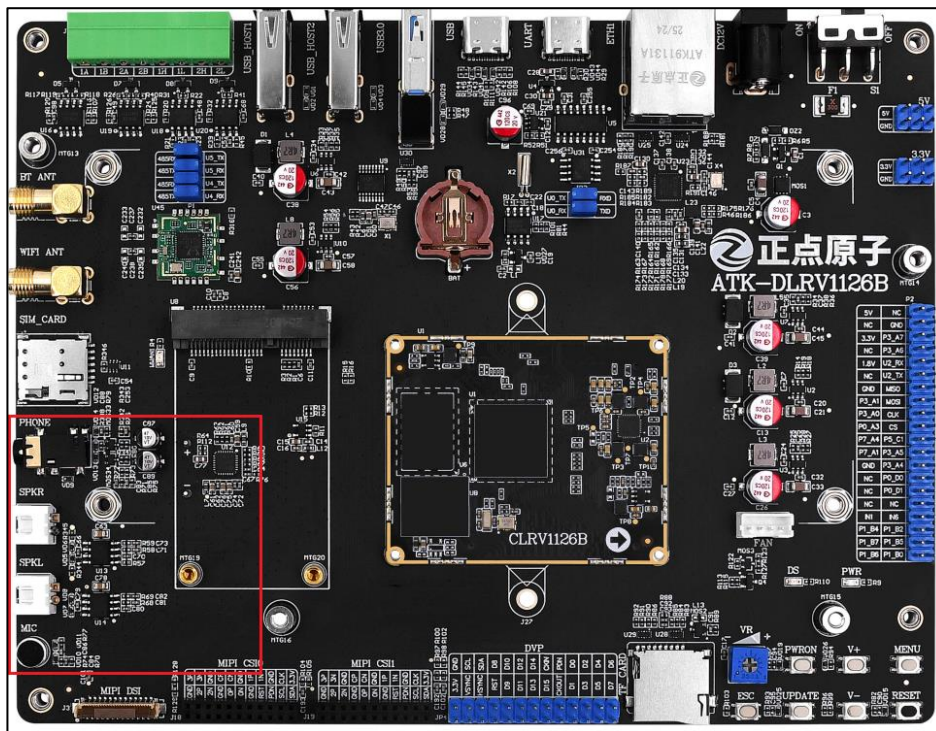


Figure 2.5-1 Audio Interface

2.5.1 Audio Playback Test

You can use the system commands `aplay` or `gst-play-1.0` to play audio files. Execute the following commands to play audio files. Users can also copy audio files to the file system and use the commands `aplay / gst-play-1.0 + audio file name` for playback.

gst-play-1.0 /userdata/piano2-CoolEdit.mp3 # Play using **gst-play-1.0**, which supports WAV, MP3, and other audio formats.

2.5.2 Recording Test

`aplay -l` # Note: it's a lowercase L, not 1!

Check the sound card number. You can see that the es8390 is sound card 0. All places where the number 0 is used below are related to this.

```
root@ATK-DLRV1126B:/# aplay -l
**** List of PLAYBACK Hardware Devices ****
card 0: rockchip-es8390 [rockchip-es8390], device 0: dailink-multicodecs ES8389 HiFi-0 [dailink-multicodecs ES8389 HiFi-0]
  Subdevices: 1/1
    Subdevice #0: subdevice #0
root@ATK-DLRV1126B:/#
```

Figure 2.5-2 Sound card list of the Buildroot system

Then we first set the recording gain of the sound card.

```
/usr/bin/amixer -c 0 cset name='ADC OSR Volume ON' 'on'
/usr/bin/amixer -c 0 set 'ADC OSR' 60%
/usr/bin/amixer -c 0 set 'ADC2DAC Mixer' 90%
/usr/bin/amixer -c 0 set 'ADCL' 90%
/usr/bin/amixer -c 0 set 'ADCL PGA' 80%
/usr/bin/amixer -c 0 set 'ADCR' 90%
/usr/bin/amixer -c 0 set 'ADCR PGA' 80%
```

Then use the following command to perform the recording test:

```
/usr/bin/arecord -r 44100 -f S16_LE -c 2 -d 5 record.wav -D plughw:0,0
```

Command Parameter Explanation:

- `/usr/bin/arecord`: ALSA audio recording tool
- `-r 44100`: Sampling rate 44.1 kHz
- `-f S16_LE`: Audio format is signed 16-bit little-endian
- `-c 2`: 2 channels
- `-d 5`: Recording duration 5 seconds
- `record.wav`: Output audio file name
- `-D plughw:0,0`: Specify the audio device

Play the audio file recorded above.

Note: The size of the generated file depends on the format and recording length set by the user.

Note: Since the capture volume is set too high during recording and the microphone has no noise reduction, there may be noise. Please lower the capture volume (sensitivity) before testing again.

```
aplay record.wav # Play the recording.
```

2.6 Camera Test

The ATK-DLRV1126B supports MIPI CSI cameras. Currently, only the IMX415 camera (8 megapixels) is supported on this development board.

The Buildroot Linux system is configured by default for the MIPI CSI0 and MIPI CSI1 interfaces. You can connect the camera module to either of these two MIPI CSI interfaces, or you can connect two cameras at the same time, as shown in the figure below:

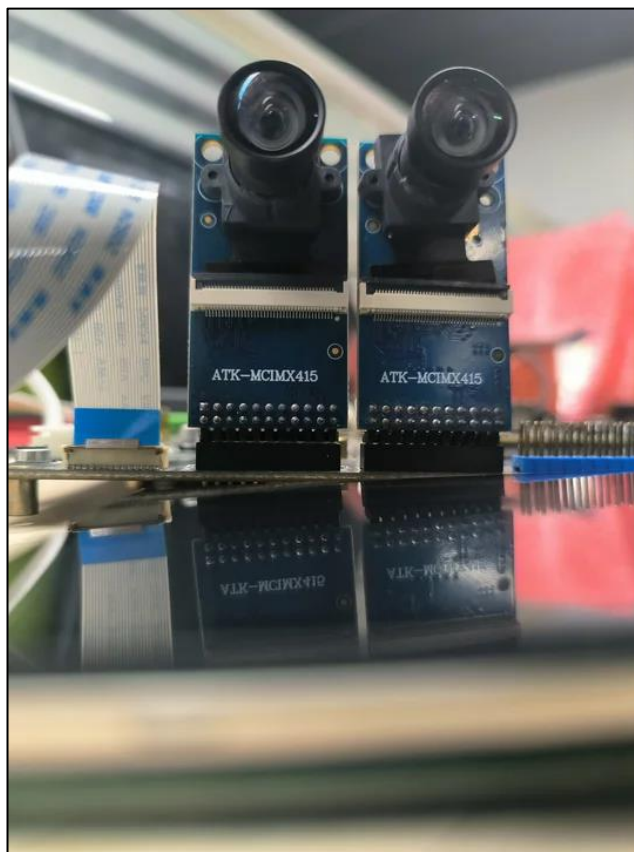


Figure 2.6-1 Connect the camera.

The corresponding relationship between the silk screen on the development board and the /dev/videoX node is shown in the table below:

Development board silk screen	/dev/videoX
MIPI CSI0	/dev/video23
MIPI CSI1	/dev/video31

After connecting the MIPI screen and MIPI camera to the development board, you can use the following commands to test the camera:

```
gst-launch-1.0 v4l2src device=/dev/video23 ! video/x-raw,format=NV12,width=640,height=480,\
framerate=30/1 ! videoconvert ! autovideosink
```

2.7 RTC Test

The ATK-DLRV1126B development board is equipped with a hardware real-time clock (RTC). The RTC clock chip AT8563T on the baseboard is an external RTC clock. To improve clock accuracy, a high-precision crystal oscillator is required.

Please check whether an RTC coin battery is installed on the baseboard. You can also use a multimeter to verify the battery voltage; a normal reading is approximately 3.3 V. This prevents time loss due to a depleted RTC battery.

(Note: Batteries are consumables. Please replace the battery if it is dead. The required battery type is CR1220 coin cell.)

The Linux system has two types of clocks: one is the **system time (software clock)**, and the other is the **hardware clock**. You can use the date and hwclock commands to view and set the system time

and hardware time respectively. The system clock will be lost when power is removed, while the RTC clock can run for a long time if a battery is installed. The system clock is synchronized with the RTC clock when the system reboots.

To view the system clock, use the date command.

Note: The factory system uses the ntpd time synchronization service. Therefore, if you connect the network when powering on and have Internet access, your system time will be synchronized with the network time.

However, this synchronized time is **not** written to the hardware clock.

As a result, you may see two different times when using date and hwclock.

Set the current system clock.

```
date -s "2025-11-20 11:16:00"
```

Check the system clock.

```
date
```

Write the system time to the hardware clock using hwclock.

```
hwclock -w
```

Check the hardware clock to verify the time set by the hwclock -w command above.

```
hwclock
```

```
root@ATK-DLRV1126B:/# date -s "2025-11-20 11:16:00"
Thu Nov 20 11:16:00 AM CST 2025
root@ATK-DLRV1126B:/# date
Thu Nov 20 11:16:02 AM CST 2025
root@ATK-DLRV1126B:/# hwclock -w
root@ATK-DLRV1126B:/# hwclock
Thu Nov 20 11:16:14 2025  0.000000 seconds
root@ATK-DLRV1126B:/#
```

Figure 2.7-1 clock

2.8 ADC Test

Note: The absolute maximum input voltage for ADC sampling is 1.8 V. Do not exceed 1.8 V, otherwise the chip may be damaged.

On the ATK-DLRV1126B baseboard, there is an ADC test channel at the position marked VR on the silk screen. The potentiometer is connected to channel 4 of ADC0, as shown in the figure below:

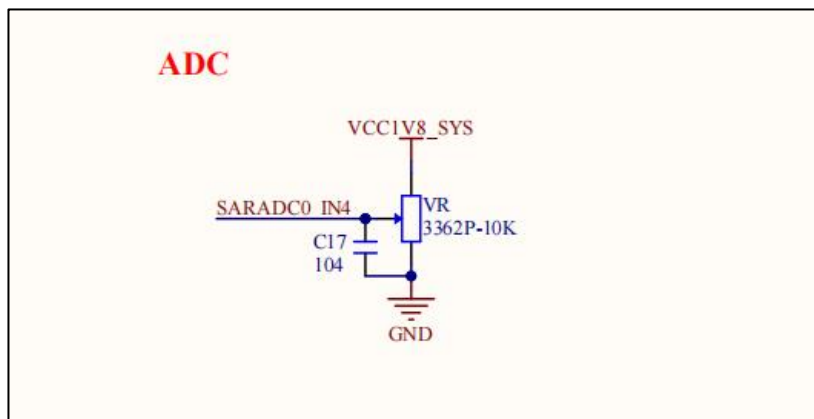


Figure 2.8-1 Potentiometer Circuit

In the Linux system, ADC acquisition is under the IIO subsystem. The ADC controller can be accessed through the interface provided by SYSFS, and it uses a 13-bit resolution.

Use the following command to view the raw data collected by the ADC. The maximum value is 2 to the power of n minus 1.

```
cat /sys/bus/iio/devices/iio\:device0/in_voltage4_raw
```

```
root@ATK-DLRV1126B:/# cat /sys/bus/iio/devices/iio\:device0/in_voltage4_raw
8191
root@ATK-DLRV1126B:/# _
```

Figure 2.8-2 Read ADC Value

Convert the AD-converted value to the desired voltage value using a standard reference voltage. The calculation formula is as follows:

$$\text{voltage} = \text{raw_value} * \text{scale_value};$$

Explanation:

raw_value: Raw data collected by the ADC.

voltage: The actual voltage value to be measured.

scale_value: Scaling factor, which equals $1800 / 8192$.

Here, 8192 is equal to 2^{13} (13 represents the ADC resolution), and 1800 is the reference voltage in millivolts (mV).

```
cat /sys/bus/iio/devices/iio\:device0/in_voltage_scale
```

```
root@ATK-DLRV1126B:/# cat /sys/bus/iio/devices/iio\:device0/in_voltage_scale
0.219726562
root@ATK-DLRV1126B:/#
```

Figure 2.8-3 Scaling Factor

Final calculation result:

$$\text{voltage} = 8189 \times 0.219726562 \approx 1800 \text{ mV}.$$

In fact, only when **raw_value** equals 8192 will **voltage** be exactly 1800 mV. This difference may be acceptable in most applications, but special attention may be required for high-precision measurement scenarios.

2.9 UART Test

The ATK-DLRV1126B development board has two RS485 serial ports. One RS485 serial port is used here for demonstration.

Connect 1A and 1B to the A and B terminals of the USB-to-RS485 module respectively, as shown in the figure below:

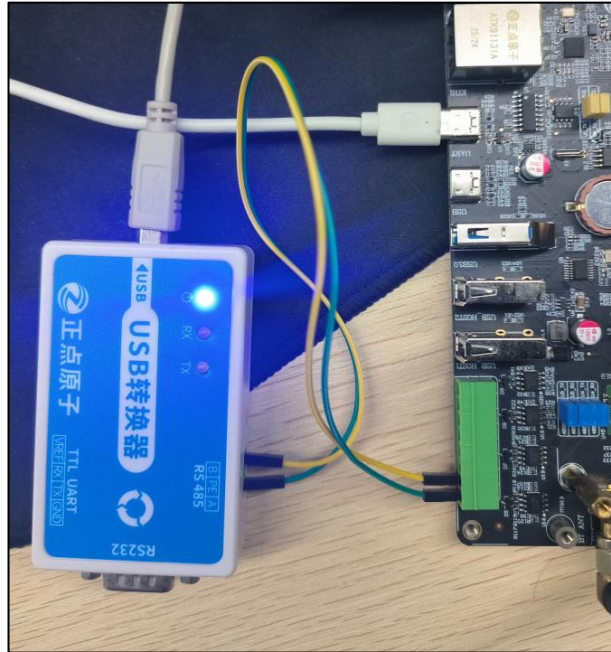


Figure 2.9-1 RS485 serial port

Run the following commands to initialize the serial port (to test the other RS485 serial port, simply replace /dev/ttyS4 with /dev/ttyS5). The prompt messages can be ignored without affecting the operation.

```
microcom -s 1500000 -X /dev/ttyS4
```

```
root@ATK-DLRV1126B:~# microcom -s 1500000 -X /dev/ttyS4
[ 183.821304] of_dma_request_slave_channel: dma-names property of node '/serial@21190000' missing or empty
[ 183.821361] dw-apb-uart 21190000.serial: failed to request DMA, use interrupt mode
```

Figure 2.9-2 Open the serial port.

The test results are as follows (you can choose any serial port testing tool you prefer):

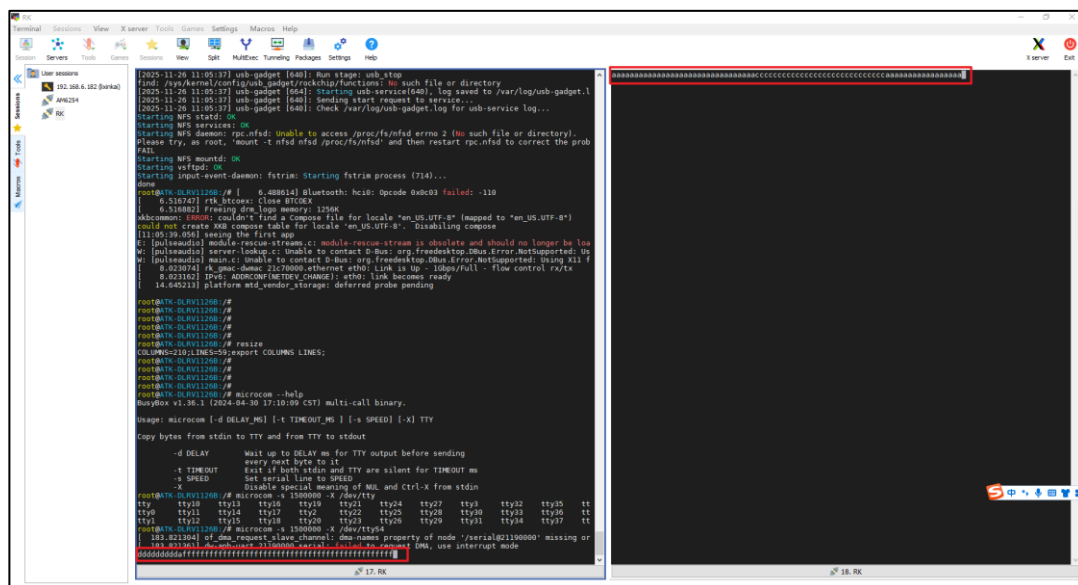


Figure 2.9-3 Serial Communication Test

2.10 CAN Test

The ATK-DLRV1126B development board is equipped with two CAN interfaces, as shown in the figure below:

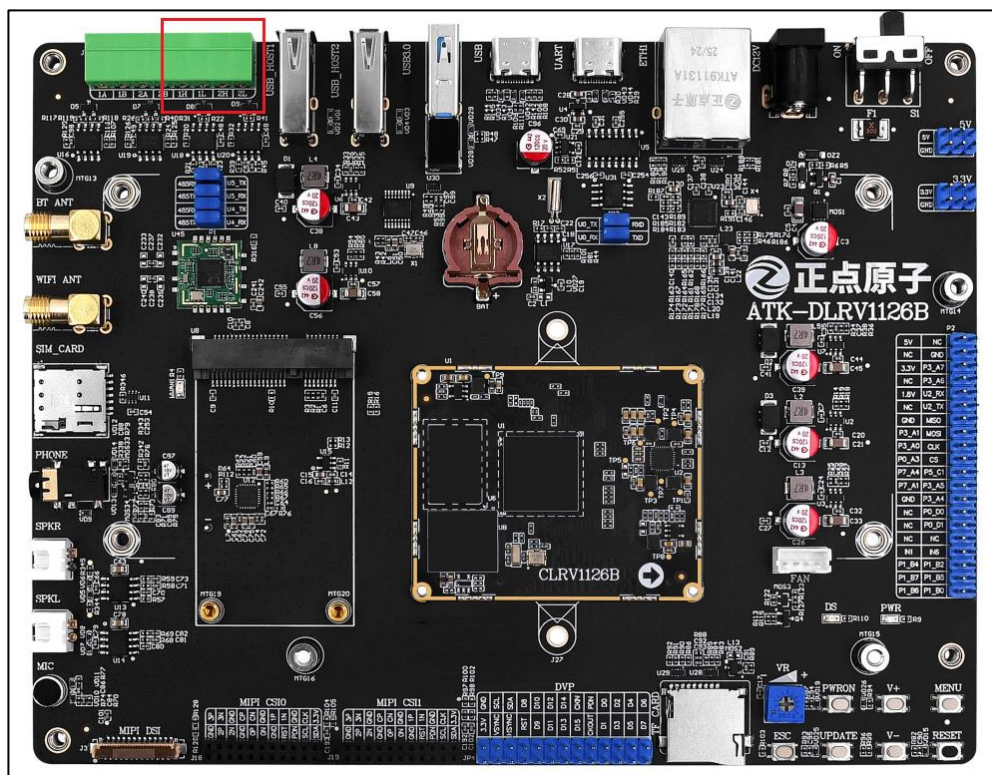


Figure 2.10-1 CAN Interface

We test by connecting the two CAN interfaces together, as shown in the following connection diagram:

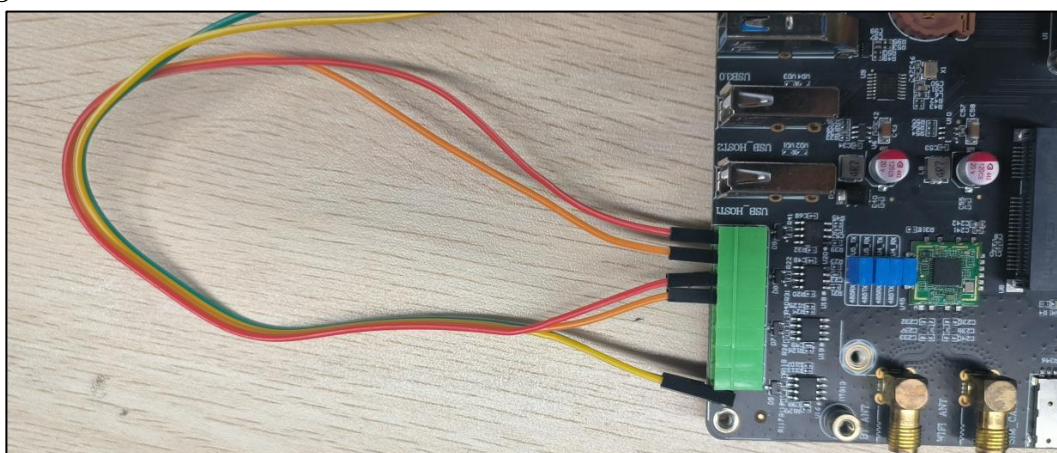


Figure 2.10-2 CANFD Connection

2.10.1 Standard CAN Test

Execute the following commands in sequence to initialize, configure and enable the CAN controller:

```
# close the interface
```

```

ip link set can0 down
ip link set can1 down

# Configure standard CAN mode, disable CANFD, and set the baud rate to 500kbps.
ip link set can0 type can bitrate 500000 fd off
ip link set can1 type can bitrate 500000 fd off

# Enable the interface
ip link set can0 up
ip link set can1 up

```

```

root@ATK-DLRV1126B:/#
root@ATK-DLRV1126B:/#
root@ATK-DLRV1126B:/#
root@ATK-DLRV1126B:/#
root@ATK-DLRV1126B:/# ip link set can0 down
root@ATK-DLRV1126B:/# ip link set can1 down
root@ATK-DLRV1126B:/# ip link set can0 type can bitrate 500000 fd off
root@ATK-DLRV1126B:/# ip link set can1 type can bitrate 500000 fd off
root@ATK-DLRV1126B:/# ip link set can0 up
root@ATK-DLRV1126B:/# ip link set can1 up
[ 84.970661] IPv6: ADDRCONF(NETDEV_CHANGE): can1: link becomes ready
root@ATK-DLRV1126B:/#

```

Figure 2.10-3 Initialize CAN Test

Execute the following command to make CAN0 receive data in the background.

```
candump can0 &
```

```

root@ATK-DLRV1126B:/#
root@ATK-DLRV1126B:/# candump can0 &
[1] 969
root@ATK-DLRV1126B:/#

```

Then execute the following command to make CAN1 send data to CAN0 continuously.

```
cangen can0 -g 100
```



```

root@ATK-DLRV1126B:/# cangen can0 -g 100
can0 674 [8] 2C 0E F0 56 FA 6B DF 0C
can0 7AA [3] 4F A6 C3
can0 7C4 [8] 8D 8C 5E 4F 26 DD E5 7D
can0 4D6 [8] 4A 3A B6 46 35 B8 AA 6B
can0 29C [8] 18 24 79 16 71 0B 03 01
can0 0F7 [7] 3B 28 C7 15 64 B1 8C
can0 40E [7] 09 60 CB 75 71 24 96
can0 236 [7] 6E 1D 41 19 AA 40 63
can0 051 [8] A4 AC 42 43 FB 0F AE 5C
can0 0AD [4] 80 1C 3C 04
can0 581 [8] 98 9D 68 33 58 32 69 64
can0 715 [2] 8D EA
can0 720 [6] 23 DD 76 7A 18 80
can0 74D [8] 7C 31 C1 47 5C A3 E9 52
can0 775 [6] CD C7 7F 12 AB C1
can0 74D [8] 55 02 56 26 9E 1F 60 71
can0 0D5 [8] 9A 2F 0E 4E 82 E9 C5 24
can0 1ED [8] F4 A9 48 5A 6F DF 9B 43
can0 527 [8] C7 11 05 28 3C 64 BB 29
can0 76E [4] EE 35 2F 23
can0 2FA [1] A7
can0 06F [3] A4 AD C6
can0 1A9 [1] 8F
can0 2AD [4] 95 E8 D5 09
can0 2DE [8] 05 55 BF 16 CB 5C 6D 62
can0 449 [8] 3A 3C 09 26 70 59 13 50
can0 685 [1] AD
can0 256 [8] 82 2C AD 11 50 55 CD 24
can0 6AD [8] 98 5F AA 69 1C 1F 88 68
can0 34D [8] 01 07 75 5A F6 F4 C4 01
can0 2AE [0]
can0 475 [3] DD 48 B8
can0 6B2 [2] 1E 14
can0 4DB [8] 6C 7C 36 76 60 E3 38 7A
can0 65A [8] 54 49 C2 00 B0 E8 04 53
can0 5B5 [6] 01 3E D2 77 62 34
can0 100 [8] 7E 53 2F 30 4D D4 93 20
can0 2D6 [8] 43 C9 58 22 84 2D A5 0E
can0 310 [1] DF
can0 5B4 [8] D8 1E DC 41 66 EC 98 5C
can0 19F [6] 62 0F BC 02 7A 8E
can0 34F [8] DB 71 4C 04 AA 21 4F 26
can0 5E8 [8] 5A 0A 54 79 9D F3 FA 75
can0 105 [8] FF 27 A2 3D 05 1A F3 71
^C can0 5F5 [8] 52 EE 86 12 CB 90 90 35
root@ATK-DLRV1126B:/#

```

2.10.2 CANFD Test

Then execute the following commands in sequence to initialize, configure and enable the CAN controller:

```

# close the interface
ip link set can0 down
ip link set can1 down

# Configure to CAN FD mode, and set the data segment baud rate to 5 Mbps.
ip link set can0 type can bitrate 500000 dbitrate 5000000 fd on
ip link set can1 type can bitrate 500000 dbitrate 5000000 fd on

# Start the interface
ip link set can0 up

```



```
ip link set can1 up
```

```
root@ATK-DLRV1126B:/#  
root@ATK-DLRV1126B:/# ip link set can0 down  
root@ATK-DLRV1126B:/# ip link set can1 down  
root@ATK-DLRV1126B:/# ip link set can0 type can bitrate 500000 dbitrate 5000000 fd on  
root@ATK-DLRV1126B:/# ip link set can1 type can bitrate 500000 dbitrate 5000000 fd on  
root@ATK-DLRV1126B:/# ip link set can0 up  
[ 807.630658] IPv6: ADDRCONF(NETDEV_CHANGE): can0: link becomes ready  
root@ATK-DLRV1126B:/# ip link set can1 up  
[ 814.525565] IPv6: ADDRCONF(NETDEV_CHANGE): can1: link becomes ready  
root@ATK-DLRV1126B:/#
```

Figure 2.10-4 CAN Initialization

Next, execute the following commands to make CAN0 receive and print data in the background:

candump can0 &

Then use the following command to let can1 send data:

cangen can1 -g 100 -I 123 -D 1122334455667788 -L 8 -f

The test results are shown in the figure below:

[illegible]

Figure 2.10-5 CAN Test

2.11 USB Test

The USB system includes 2 USB 2.0 HOST ports, 1 USB 3.0 HOST port, and 1 USB 2.0 Device Type-C port.

(The USB 3.0 HOST and USB 2.0 Device belong to the same USB controller and cannot be used with devices inserted simultaneously.)

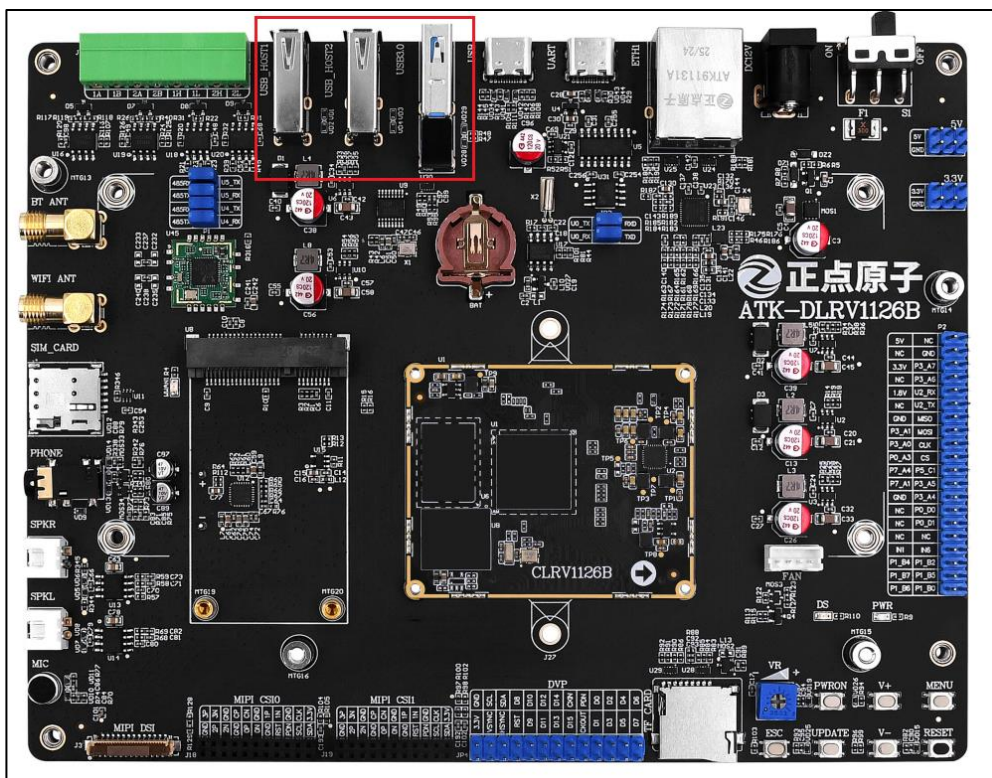


Figure 2.11-1 USB Interface

2.11.1 USB HOST Test

Insert a USB flash drive formatted in FAT32 directly into any USB port on the ATK-DLRV1126B development board using a card reader.

```

root@ATK-DLRV1126B:/# [10364.028178] usb 1-1.4: new high-speed USB device number 4 using ehci-platform
[10364.137547] usb 1-1.4: New USB device found, idVendor=14cd, idProduct=1212, bcdDevice= 1.00
[10364.137571] usb 1-1.4: New USB device strings: Mfr=1, Product=3, SerialNumber=2
[10364.137578] usb 1-1.4: Product: Mass Storage Device
[10364.137584] usb 1-1.4: Manufacturer: Generic
[10364.137591] usb 1-1.4: SerialNumber: 121220160204
[10364.139315] usb-storage 1-1.4:1.0: USB Mass Storage device detected
[10364.139953] scsi host0: usb-storage 1-1.4:1.0
[10365.153146] scsi 0:0:0:0: Direct-Access   Mass        Storage Device 1-00:00:0-0-ANCSI-0 CCS
[10365.259443] sd 0:0:0:0: [sda] 61120512 512-byte logical blocks: (31.3 GB/29.1 GiB) 容量大小
[10365.260955] sd 0:0:0:0: [sda] Write Protect is off
[10365.260978] sd 0:0:0:0: [sda] No Caching mode page found
[10365.260987] sd 0:0:0:0: [sda] Assuming drive cache: write through
[10365.266207] sda: sda1 分区
[10365.266598] sd 0:0:0:0: [sda] Attached SCSI removable disk
[10365.491372] FAT-fs (sda1): utf8 is not a recommended IO charset for FAT filesystems, filesystem will be case sensitive!
[10365.496152] FAT-fs (sda1): Volume was not properly unmounted. Some data may be corrupt. Please run fsck.

root@ATK-DLRV1126B:/#
root@ATK-DLRV1126B:/# df -h
Filesystem      Size  Used Avail Use% Mounted on
/dev/root        5.0G  924M  4.8G  17% /
devtmpfs        457M   0  457M   0% /dev
tmpfs           485M  540K  484M   1% /run
tmpfs           485M  220K  485M   1% /var/log
tmpfs           485M   0  485M   0% /dev/shm
tmpfs           485M  240K  485M   1% /tmp
/dev/mmcblk0p8  938M  332K  937M   1% /userdata
/dev/mmcblk0n7  124M   12M  112M  10% /oem
/dev/sda1       30G  128K   30G   1% /mnt/udisk 自动挂载路径
root@ATK-DLRV1126B:/#

```

Figure 2.11-2 USB Drive Print Information

We can directly enter the /mnt/udisk directory to perform file read and write operations.

Note: The test read/write speed is related to the maximum read/write speed of your card reader, USB flash drive, or TF card.

Read Speed Test:

```
hdparm -t /dev/sda1
```

Write Speed Test:

```
time dd if=/dev/zero of=/mnt/udisk/test bs=1024k count=100 conv=fdatasync
```

```
rm /mnt/udisk/test
```

This represents writing a 100MiB file, with a write speed of 9.0 MB/s. In practice, the larger the file being written, the closer the average write speed will be to the actual value.

```
root@ATK-DLRV1126B:/# hdparm -t /dev/sda1

/dev/sda1:
Timing buffered disk reads: 52 MB in 3.05 seconds = 17441 kB/s
root@ATK-DLRV1126B:/# time dd if=/dev/zero of=/mnt/udisk/test bs=1024K count=100 conv=fdatasync
100+0 records in
100+0 records out
104857600 bytes (105 MB, 100 MiB) copied, 11.6085 s, 9.0 MB/s

real    0m11.725s
user    0m0.000s
sys     0m0.573s
root@ATK-DLRV1126B:/#
```

Figure 2.11-3 USB disk test

2.11.2 USB TypeC Slave Test

The USB TypeC interface on the ATK-DLRV1126B supports the ADB function. When connected, it can be recognized as an ADB device, acting as a slave device.

2.12 Network Test

The ATK-DLRV1126B development board is equipped with a gigabit Ethernet chip, supporting auto-negotiation for 10/100/1000M. Use the following commands in the file system to test the network.

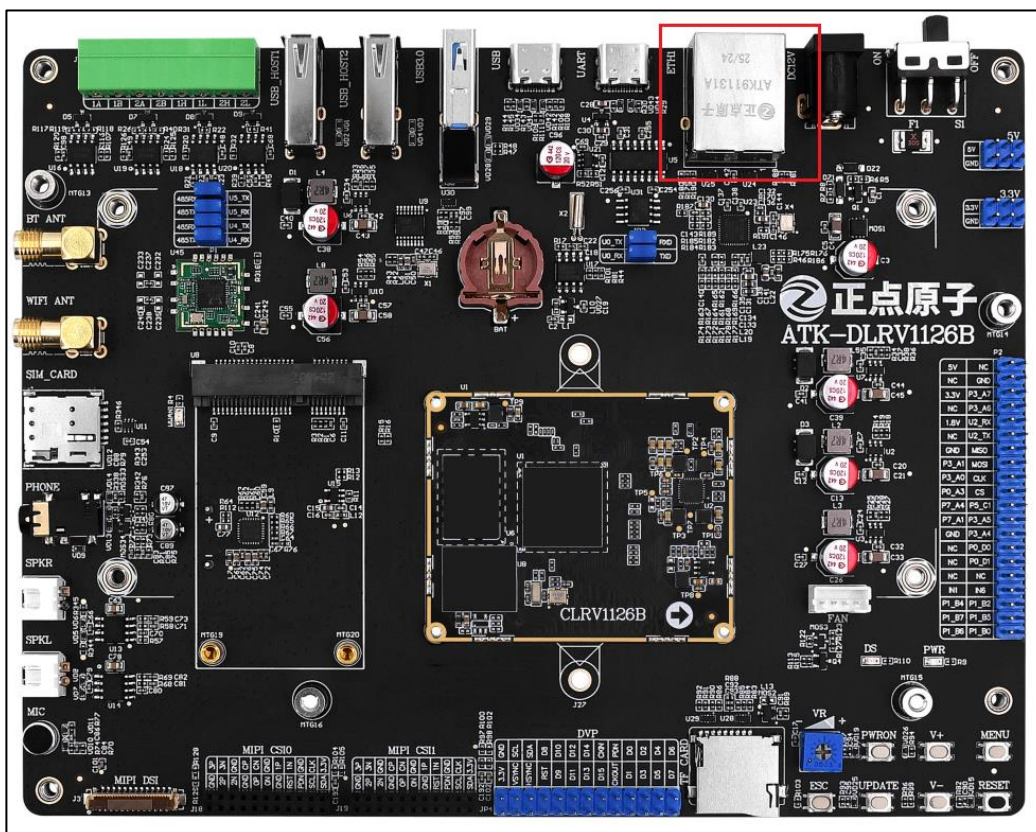


Figure 2.12-1 Onboard Gigabit Ethernet Port

Connect the Ethernet cable to the port and plug the other end into a router to ensure internet access. The default factory system obtains an IP address dynamically via DHCP.

Check the acquired network IP address.

ifconfig

```
root@ATK-DLRV1126B:/# [11042.850537] rk_gmac-dwmac 21c70000.ethernet eth0: Link is Up - 1Gbps/Full - flow control rx/tx
[11042.850608] IPv6: ADDRCONF(NETDEV_CHANGE): eth0: link becomes ready

root@ATK-DLRV1126B:/#
root@ATK-DLRV1126B:/#
root@ATK-DLRV1126B:/# ifconfig
root@ATK-DLRV1126B:/#
eth0      Link encap:Ethernet  HWaddr 36:AD:63:E2:79:C4
          inet addr:192.168.6.124  Bcast:192.168.6.255  Mask:255.255.255.0
          inet6 addr: fe80::34ad:63ff:fe27:9c4/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:22  errors:0  dropped:4  overruns:0  frame:0
          TX packets:17  errors:0  dropped:0  overruns:0  carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:2627 (2.5 KiB)  TX bytes:2513 (2.4 KiB)
          Interrupt:79

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:65536  Metric:1
          RX packets:11738  errors:0  dropped:0  overruns:0  frame:0
          TX packets:11738  errors:0  dropped:0  overruns:0  carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:704252 (687.7 KiB)  TX bytes:704252 (687.7 KiB)

root@ATK-DLRV1126B:/#
```

Figure 2.12-2 Obtain an IP address automatically.

Check whether the network can access the Internet. If there is a data reply, the network function is normal. Press Ctrl+c to terminate the command.

ping www.baidu.com

```
root@ATK-DLRV1126B:/# ping www.baidu.com
PING www.baidu.com (183.2.172.17) 56(84) bytes of data.
64 bytes from 183.2.172.17: icmp_seq=1 ttl=52 time=6.80 ms
64 bytes from 183.2.172.17: icmp_seq=2 ttl=52 time=6.68 ms
64 bytes from 183.2.172.17: icmp_seq=3 ttl=52 time=6.78 ms
64 bytes from 183.2.172.17: icmp_seq=4 ttl=52 time=6.12 ms
64 bytes from 183.2.172.17: icmp_seq=5 ttl=52 time=6.26 ms
64 bytes from 183.2.172.17: icmp_seq=6 ttl=52 time=6.17 ms
64 bytes from 183.2.172.17: icmp_seq=7 ttl=52 time=6.25 ms
64 bytes from 183.2.172.17: icmp_seq=8 ttl=52 time=7.02 ms
64 bytes from 183.2.172.17: icmp_seq=9 ttl=52 time=6.15 ms
64 bytes from 183.2.172.17: icmp_seq=10 ttl=52 time=6.16 ms
64 bytes from 183.2.172.17: icmp_seq=11 ttl=52 time=6.99 ms
64 bytes from 183.2.172.17: icmp_seq=12 ttl=52 time=6.85 ms
^C
--- www.baidu.com ping statistics ---
12 packets transmitted, 12 received, 0% packet loss, time 11017ms
rtt min/avg/max/mdev = 6.117/6.519/7.022/0.345 ms
root@ATK-DLRV1126B:/#
```

Figure 2.12-3 Ping Test

Test whether it is a Gigabit network. Note that a Gigabit Ethernet cable, a Gigabit router or a Gigabit switch must be used, and the PC network card must also be a Gigabit network card. Otherwise, the tested or recognized speed may be 100M! As shown in the figure below, if it displays 1Gbps/Full, it indicates a Gigabit network.

```
root@ATK-DLRV1126B:/#
root@ATK-DLRV1126B:/# [11042.850537] rk_gmac-dwmac 21c70000.ethernet eth0: Link is Up - 1Gbps/Full - flow control rx/tx
[11042.850608] IPv6: ADDRCONF(NETDEV_CHANGE): eth0: link becomes ready
```

Figure 2.12-4 Print information when the development board is connected to the network

The iperf3 command can be used to test the connection speed of the Gigabit network.

Set Ubuntu as the server. Ubuntu and the development board must be on the same LAN. If the iperf3 command is not available on Ubuntu, install it by entering `sudo apt-get install iperf3`.

Check the Ubuntu IP address.

ifconfig


```

lixinkai@lixinkai-virtual-machine:~$ ifconfig
ens33: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.6.182 netmask 255.255.255.0 broadcast 192.168.6.255
    inet6 fe80::769a:692d:c7e0:629a prefixlen 64 scopeid 0x20<link>
    ether 00:50:56:31:79:59 txqueuelen 1000 (以太网)
    RX packets 411237 bytes 129026165 (129.0 MB)
    RX errors 0 dropped 27434 overruns 0 frame 0
    TX packets 2738375 bytes 4085118552 (4.0 GB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (本地环回)
    RX packets 37265 bytes 23873746 (23.8 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 37265 bytes 23873746 (23.8 MB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

```

Figure 2.12-5 Ubuntu IP address

Enter `iperf3 -s` to set Ubuntu as the server.

```
iperf3 -s
```

```

lixinkai@lixinkai-virtual-machine:~$ iperf3 -s
-----
Server listening on 5201
-----

```

Figure 2.12-6 Iperf3 Service

The ATK-DLRV1126B acts as the client. Just enter the IP address of the server (Ubuntu) using the `iperf3` command.

```
iperf3 -c 192.168.6.182 -i 1
```

Command Explanation:

- `-s`: Set the current device as the server.
- `-c`: Set the current device as the client.
- **192.168.6.182**: IP address of the server.
- `-i`: Interval for periodic output.

```

root@ATK-DLRV1126B:/# iperf3 -c 192.168.6.182 -i 1
Connecting to host 192.168.6.182, port 5201
[ 5] local 192.168.6.124 port 46650 connected to 192.168.6.182 port 5201
[ ID] Interval      Transfer    Bitrate      Retr  Cwnd
[ 5] 0.00-1.00    sec    113 MBytes  944 Mbits/sec    0   556 KBytes
[ 5] 1.00-2.00    sec    112 MBytes  940 Mbits/sec    0   556 KBytes
[ 5] 2.00-3.00    sec    112 MBytes  942 Mbits/sec    0   556 KBytes
[ 5] 3.00-4.00    sec    112 MBytes  938 Mbits/sec    0   584 KBytes
[ 5] 4.00-5.00    sec    113 MBytes  946 Mbits/sec    0   584 KBytes
[ 5] 5.00-6.00    sec    112 MBytes  939 Mbits/sec    0   584 KBytes
[ 5] 6.00-7.00    sec    113 MBytes  944 Mbits/sec    0   584 KBytes
[ 5] 7.00-8.00    sec    112 MBytes  942 Mbits/sec    0   584 KBytes
[ 5] 8.00-9.00    sec    112 MBytes  939 Mbits/sec    0   584 KBytes
[ 5] 9.00-10.00   sec    112 MBytes  939 Mbits/sec    0   584 KBytes
-----
[ ID] Interval      Transfer    Bitrate      Retr
[ 5] 0.00-10.00   sec    1.10 GBytes  941 Mbits/sec    0
[ 5] 0.00-10.04   sec    1.09 GBytes  936 Mbits/sec
sender
receiver
iperf Done.
root@ATK-DLRV1126B:/#

```

Figure 2.12-7 Iperf3 Test

A normal Gigabit network speed should show a Bitrate of approximately 940 Mbits/sec for both transmit and receive. Please do not run other CPU-intensive applications before the test to avoid affecting the test speed. If not, check whether you are using a Gigabit Ethernet cable, whether the PC network adapter is a Gigabit adapter, and whether the connected router is a Gigabit router. There must be no 100M equipment in the link; otherwise, the test result will be a 100M network.

2.13 TF Card & EMMC Test

Command Notes:

The time command is often used to measure the execution time of a command. dd is used for copying: it reads from the if (input file) and writes to the of (output file). bs represents the block size per write operation, and count is the number of blocks to read and write.

if=/dev/zero generates no I/O operations and can continuously output data, so it is used to test pure write speed.

2.13.1 TF Card Test

After the development board boots up, insert the TF card into the card slot on the baseboard of the development board. The print information is as follows.

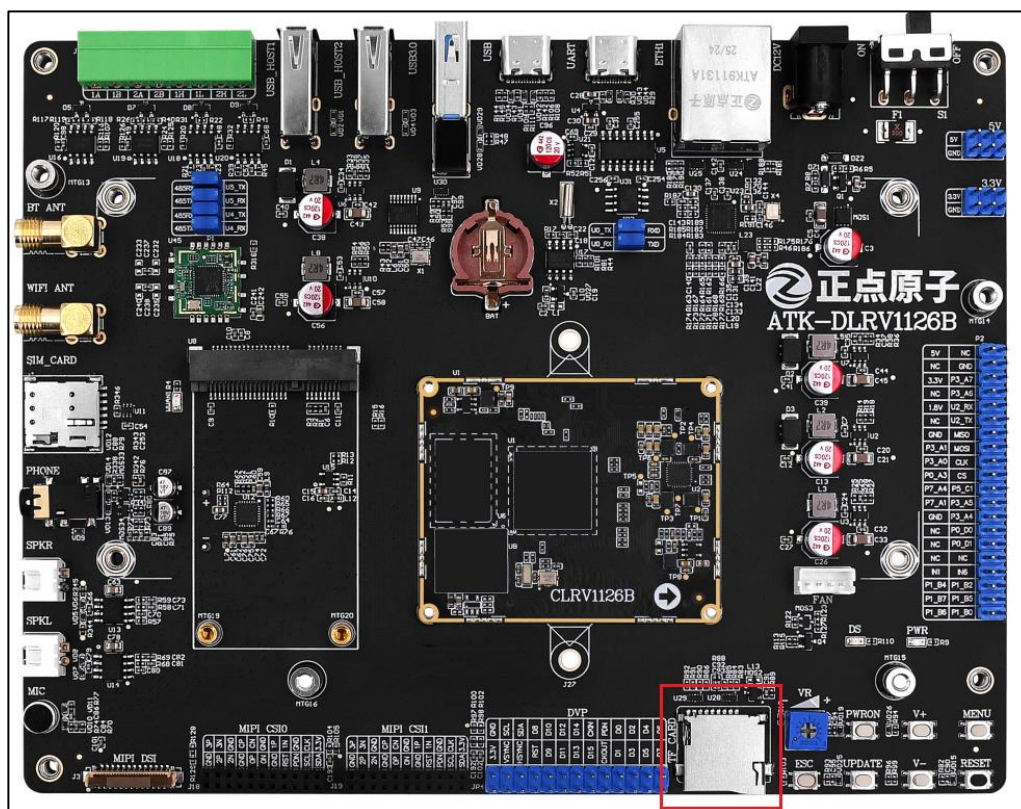


Figure 2.13-1 TF Card Slot

The development board will automatically mount the TF card. (Note: A FAT32-formatted TF card is required.)

Use the `df -h` command to check the mounted SD partition. The running result is shown below:


```

root@ATK-DLRV1126B:/# [12005.753371] mmc_host mmc1: Bus speed (slot 0) = 198000000Hz (slot req 200000000Hz, actual 198000000Hz div = 0)
[12005.841525] dwmmc_rockchip 21d60000.mmc: Successfully tuned phase to 3
[12005.841603] mmc1: new ultra high speed SDHCI card at address 0001
[12005.842427] mmcblk1: mmc1:0001 SD32G 29.1 GiB
[12006.065685] mmc_host mmc1: Bus speed (slot 0) = 400000Hz (slot req 400000Hz, actual 400000Hz div = 0)
[12006.129263] mmc_host mmc1: Bus speed (slot 0) = 198000000Hz (slot req 200000000Hz, actual 198000000Hz div = 0)
[12006.220980] dwmmc_rockchip 21d60000.mmc: Successfully tuned phase to 135
[12006.221753] mmcblk1: p1
[12006.354486] FAT-fs (mmcblk1p1): utf8 is not a recommended IO charset for FAT filesystems, filesystem will be case sensitive!
[12006.357462] FAT-fs (mmcblk1p1): Volume was not properly unmounted. Some data may be corrupt. Please run fsck.

root@ATK-DLRV1126B:/#
root@ATK-DLRV1126B:/#
root@ATK-DLRV1126B:/# df -h
Filesystem      Size  Used Avail Use% Mounted on
/dev/root        5.9G  924M  4.8G   17% /
devtmpfs        457M   0  457M   0% /dev
tmpfs           485M  532K  484M   1% /run
tmpfs           485M  284K  484M   1% /var/log
tmpfs           485M   0  485M   0% /dev/shm
tmpfs           485M  240K  485M   1% /tmp
/dev/mmcblk0p8  938M  332K  937M   1% /userdata
/dev/mmcblk0p7  124M   12M  112M  10% /oem
/dev/mmcblk1p1  30G  101M   30G   1% /mnt/sdcard
root@ATK-DLRV1126B:/#

```

Figure 2.13-2 TF Card Print Information

For read speed testing, common commands for testing hard disk read speed include hdparm and dd. This demonstration only uses hdparm to test the read speed of the TF card. Since the data sampling time is short and the data volume is small, the speed is for reference only.

```
hdparm -t /dev/mmcblk1p1
```

```

root@ATK-DLRV1126B:/# hdparm -t /dev/mmcblk1p1
/dev/mmcblk1p1:
Timing buffered disk reads: 253 MB in 3.00 seconds = 86135 kB/s

```

Figure 2.13-3 TF Card Read Speed Test

It can be seen that the read speed is 86135 kB/s.

Write Speed Test

```
time dd if=/dev/zero of=/mnt/sdcard/test bs=1024k count=50 conv=fdatasync
```

```

root@ATK-DLRV1126B:/# time dd if=/dev/zero of=/mnt/sdcard/test bs=1024k count=50 conv=fdatasync
50+0 records in
50+0 records out
52428800 bytes (52 MB, 50 MiB) copied, 3.45352 s, 15.2 MB/s

real    0m3.467s
user    0m0.000s
sys     0m0.232s
root@ATK-DLRV1126B:/#

```

Figure 2.13-4 TF Card Write Speed Test

The dd command is used for reading. Due to the Linux system mechanism, it is generally unnecessary to deliberately release the used cache. These cache contents can improve file read and write speeds.

Run the following command to clear the cache.

```

echo 3 > /proc/sys/vm/drop_caches          # Clear cache
time dd if=/mnt/sdcard/test of=/dev/null bs=1024k  # Read Speed Test
rm /mnt/sdcard/test                          # Delete this file after reading.

```

```
root@ATK-DLRV1126B:/# echo 3 > /proc/sys/vm/drop_caches
[12173.971028] sh (756): drop_caches: 3
root@ATK-DLRV1126B:/# time dd if=/mnt/sdcard/test of=/dev/null bs=1024k
50+0 records in
50+0 records out
52428800 bytes (52 MB, 50 MiB) copied, 0.593613 s, 88.3 MB/s

real    0m0.606s
user    0m0.001s
sys     0m0.076s
root@ATK-DLRV1126B:/# rm /mnt/sdcard/test
root@ATK-DLRV1126B:/#
```

Figure 2.13-5 TF Card Read Speed Test

It can be seen that the read speed is 88.3 MB/s. The speed is for reference only; the larger the read file, the more accurate the speed.

2.13.2 EMMC Test

As a storage device, eMMC stores the Linux system. Users' future applications will be stored on this device (if no external storage is connected).

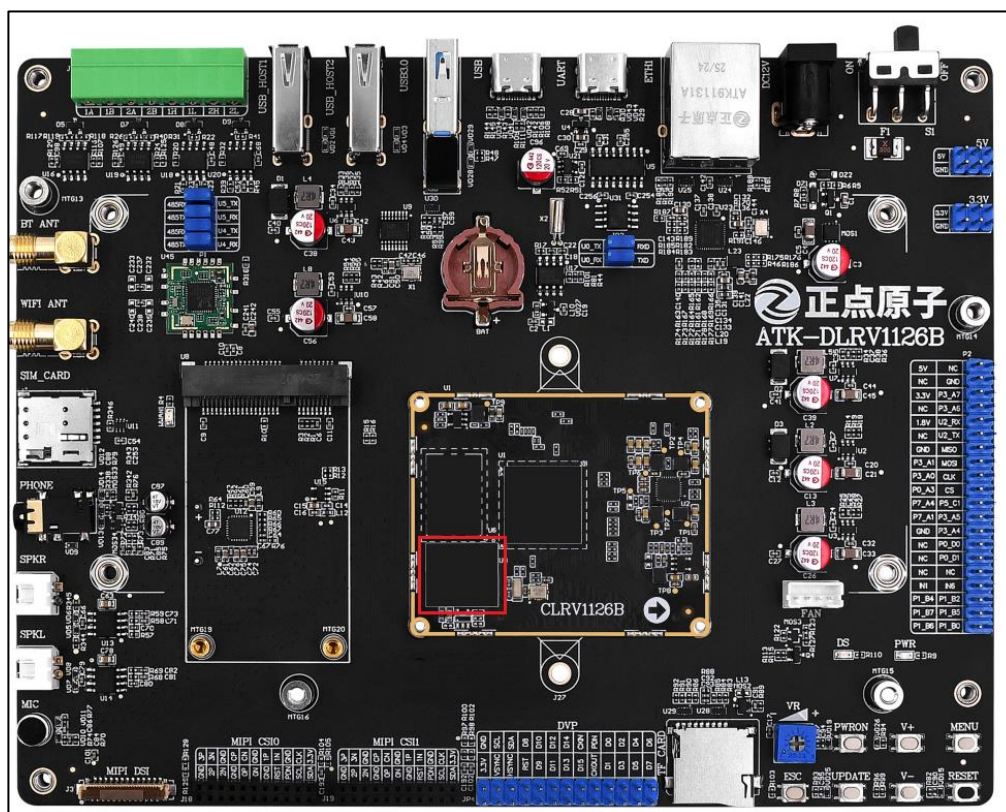


Figure 2.13-6 Onboard eMMC

Use the `df -h` command to identify the mounted eMMC partition. The result is shown below.

In fact, the eMMC storage is divided into multiple partitions, and the partition mounted to the root directory "/" is also an eMMC partition. Here we take the `mmcblk0p8` partition mounted to the `/userdata` directory as an example.

```
df
```

```

root@ATK-DLRV1126B:/# df
Filesystem      1K-blocks    Used Available Use% Mounted on
/dev/root        6159808  945480   4935776  17% /
devtmpfs         467548      0    467548   0% /dev
tmpfs            495900     536    495364   1% /run
tmpfs            495900     144    495756   1% /var/log
tmpfs            495900      0    495900   0% /dev/shm
tmpfs            495900     240    495660   1% /tmp
/dev/mmcblk0p8   960388     364    943232   1% /userdata
/dev/mmcblk0p7  126396    11400   111492  10% /oem

```

Figure 2.13-7 eMMC Partition Information

Read Speed Test

```
hdparm -t /dev/mmcblk0p8
```

```

root@ATK-DLRV1126B:/# hdparm -t /dev/mmcblk0p8
/dev/mmcblk0p8:
Timing buffered disk reads: 495 MB in 3.00 seconds = 168678 kB/s
root@ATK-DLRV1126B:/#

```

Figure 2.13-8 eMMC Read Speed Test

It can be seen that the read speed is 168678 kB/s.

Write Speed Test

```
time dd if=/dev/zero of=/userdata/test bs=1024k count=50 conv=fdatasync
```

```

root@ATK-DLRV1126B:/# time dd if=/dev/zero of=/userdata/test bs=1024k count=50 conv=fdatasync
50+0 records in
50+0 records out
52428800 bytes (52 MB, 50 MiB) copied, 0.616193 s, 85.1 MB/s

real    0m0.623s
user    0m0.004s
sys     0m0.193s
root@ATK-DLRV1126B:/#

```

Figure 2.13-9 eMMC Write Speed Test

It can be seen that the write speed is 85.1 MB/s.

The dd command is used for reading. Due to the Linux system mechanism, it is generally unnecessary to deliberately release the used cache. These cache contents can improve file read and write speeds.

Run the following command to clear the cache.

```

echo 3 > /proc/sys/vm/drop_caches          # Clear Cache
time dd if=/userdata/test of=/dev/null bs=1024k    # Read Speed Test
rm /userdata/test                             # Delete this file after reading.

```

```

root@ATK-DLRV1126B:/# echo 3 > /proc/sys/vm/drop_caches
[ 150.266867] sh (731): drop_caches: 3
root@ATK-DLRV1126B:/# time dd if=/userdata/test of=/dev/null bs=1024k
50+0 records in
50+0 records out
52428800 bytes (52 MB, 50 MiB) copied, 0.304623 s, 172 MB/s

real    0m0.317s
user    0m0.000s
sys     0m0.074s
root@ATK-DLRV1126B:/#
root@ATK-DLRV1126B:/# rm /userdata/test
root@ATK-DLRV1126B:/#

```

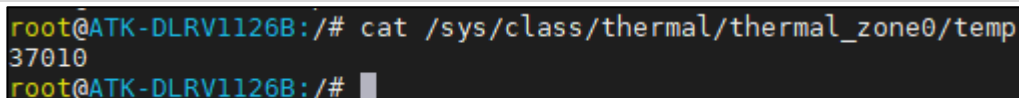
Figure 2.13-10 eMMC Write Speed Test

It can be seen that the read speed is 172 MB/s. The read file is too small, so the speed is for reference only.

2.14 Main Chip Temperature

Use the cat command to view the main chip temperature. An interface is provided at the user layer, and the displayed value is in millidegrees Celsius, as shown in the figure below.

```
cat /sys/class/thermal/thermal_zone0/temp # View Main Chip Temperature
```



```
root@ATK-DLRV1126B:/# cat /sys/class/thermal/thermal_zone0/temp
37010
root@ATK-DLRV1126B:/#
```

Figure 2.14-1 CPU Temperature

37010 means **37.01°C**.

2.15 CPU Frequency

The commands are shown below:

View Current CPU Frequency

```
cat /sys/devices/system/cpu/cpu*/cpufreq/cpuinfo_cur_freq
```

View the last set CPU frequency in software.

```
cat /sys/devices/system/cpu/cpu*/cpufreq/scaling_cur_freq
```

View supported CPU frequencies.

```
cat /sys/devices/system/cpu/cpufreq/policy0/scaling_available_frequencies
```

Enable CPU performance mode and run at maximum frequency.

```
echo performance > /sys/devices/system/cpu/cpufreq/policy0/scaling_governor
```

Switch the governor to userspace.

```
echo userspace > /sys/devices/system/cpu/cpufreq/policy0/scaling_governor
```

To set the CPU frequency, you need to switch to **userspace** first.

```
echo 1296000 > /sys/devices/system/cpu/cpufreq/policy0/scaling_setspeed
```

Then check the current running CPU frequency.

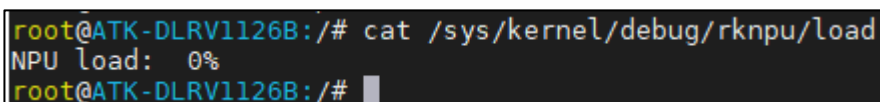
```
cat /sys/devices/system/cpu/cpu*/cpufreq/cpuinfo_cur_freq
```

2.16 View NPU Information

View NPU usage.

Note: The usage will only change when running a demo that invokes the NPU.

```
cat /sys/kernel/debug/rknpu/load
```



```
root@ATK-DLRV1126B:/# cat /sys/kernel/debug/rknpu/load
NPU load: 0%
root@ATK-DLRV1126B:/#
```

Figure 2.16-1 NPU Load

2.17 Onboard USB WIFI Test

The ATK-DLRV1126B is equipped with an **RTL8733BU** chip, which is a **WIFI + Bluetooth 2-in-1 module**.

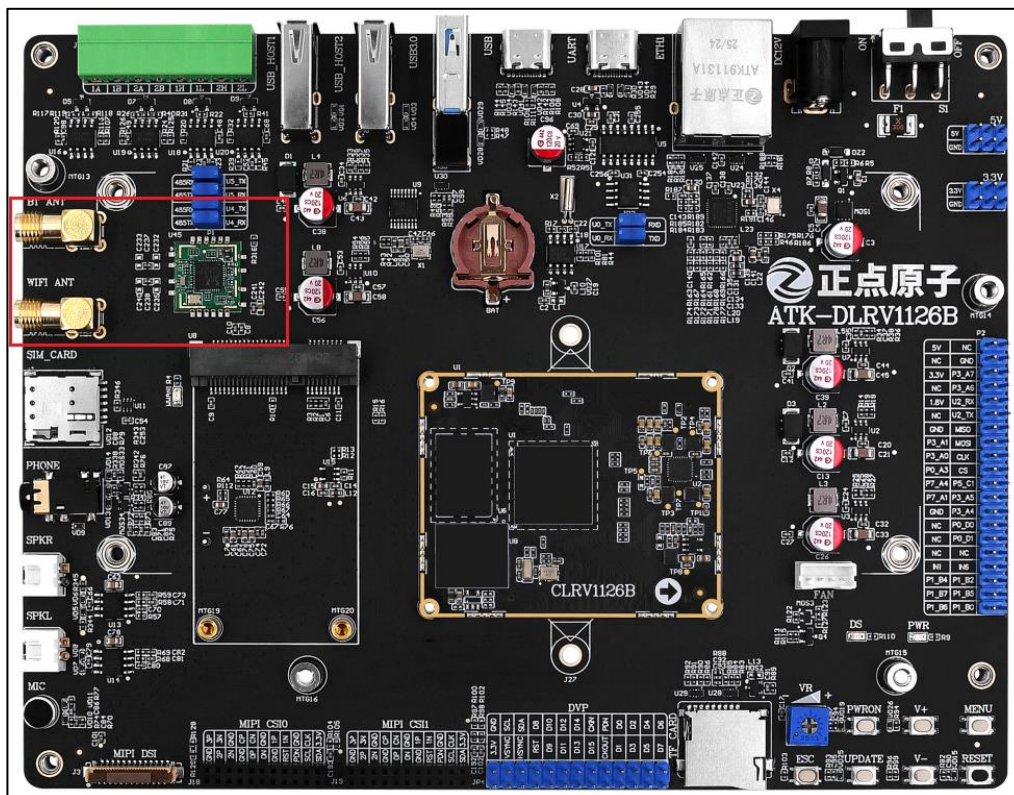


Figure 2.17-1 Onboard RTL8733BU Module

2.17.1 Station Mode

WiFi Station mode is an operating mode that allows a WiFi device to connect to a WiFi network and communicate with other devices. In Station mode, the WiFi device acts as a client to connect to an existing WiFi network, which can be a wireless network provided by a home, office, or public area. After connecting to the WiFi network, the device can perform data transmission and communication with other devices through this network.

Generally, Station mode is suitable for scenarios where the device needs to connect to an existing WiFi network, such as mobile phones, computers, smart home devices, etc.

Note: By default, wlan0 is disabled.

```
rfkill unblock wlan # Unlock the WiFi radio frequency
ifconfig           # Check whether wlan0 is enabled.
```



```

root@ATK-DLRV1126B:/# rfkill unblock wlan
root@ATK-DLRV1126B:/# [ 1312.135197] start_addr=(0x0), end_addr=(0x20000), buffer_size=(0x20000), smp_number_max=(16384)

root@ATK-DLRV1126B:/# ifconfig
eth0      Link encap:Ethernet  HWaddr 36:AD:63:E2:79:C4
          UP BROADCAST MULTICAST  MTU:1500  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)
          Interrupt:79

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:65536  Metric:1
          RX packets:1946 errors:0 dropped:0 overruns:0 frame:0
          TX packets:1946 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:116796 (114.0 KiB)  TX bytes:116796 (114.0 KiB)

wlan0     Link encap:Ethernet  HWaddr 4C:A3:8F:7B:40:BA
          UP BROADCAST MULTICAST  MTU:1500  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

root@ATK-DLRV1126B:/#

```

Figure 2.17-2 wlan network adapter

We can use the following commands to connect to WiFi.

connmanctl	# Enter the WIFI operation terminal.
connmanctl> enable wifi	# Enable WIFI. It is enabled by default.
connmanctl> agent on	# Enable proxy
connmanctl> scan wifi	# Start WIFI scanning; you can scan multiple times.
connmanctl> services	# List the scanned WiFi list.

The command execution results are shown below:

```

root@ATK-DLRV1126B:/# connmanctl
connmanctl> enable wifi
wifi is already enabled
connmanctl> scan wifi
connmanctl> services

ALIENTEK-HYS      wifi_4ca38f7b40ba_hidden_managed_psk
alientek-linux    wifi_4ca38f7b40ba_414c49454e54454b2d485953_managed_psk
alientek-linux    wifi_4ca38f7b40ba_616c69656e74656b2d6c696e7578_managed_psk
alientek-linux Wi-Fi5 wifi_4ca38f7b40ba_616c69656e74656b2d6c696e75785f57692d466935_managed_psk
ALIENTEK-17机房_5G wifi_4ca38f7b40ba_414c49454e54454b2d3137e69cbae688bf5f3547_managed_psk
ALIENTEK-17机房    wifi_4ca38f7b40ba_414c49454e54454b2d3137e69cbae688bf_managed_psk
xiaoxiao_5G       wifi_4ca38f7b40ba_7869616f7869616f5f3547_managed_psk
Xiaozhi-046D      wifi_4ca38f7b40ba_5869616f7a68692d30343644_managed_none
ALIENTEK-WX       wifi_4ca38f7b40ba_414c49454e54454b2d5758_managed_psk

Scan completed for wifi
connmanctl>
connmanctl>

```

Figure 2.17-3 WiFi List

From the scanned WiFi list, use connect XXXX to establish a connection, enter the password, and press Enter to confirm. For example, if you want to connect to "ALIENTEK-HYS" (if using an iOS personal hotspot, you may need to select **Maximize Compatibility** when turning on the hotspot. Also, note that connection failures may occur if the WiFi signal is weak (too far away) or if the password is entered incorrectly. Please confirm these two factors during testing).

The command and its running result are as follows:

connect wifi_4ca38f7b40ba_414c49454e54454b2d485953_managed_psk	# Connect
exit	# Exit connmanctl

```

connmanctl>
connmanctl> connect wifi_4ca38f7b40ba_414c49454e54454b2d485953_managed_psk
Agent RequestInput wifi_4ca38f7b40ba_414c49454e54454b2d485953_managed_psk
  Passphrase = [ Type=psk, Requirement=mandatory, Alternates=[ WPS ] ]
  WPS = [ Type=wpspin, Requirement=alternate ]
Passphrase? Enter the password
connmanctl> [ 2214.427881] IPv6: ADDRCONF(NETDEV_CHANGE): wlan0: link becomes ready
Connected wifi_4ca38f7b40ba_414c49454e54454b2d485953_managed_psk
connmanctl> exit
root@ATK-DLRV1126B:/# Connection successful

```

Figure 2.17-4 Connect to WiFi

After a successful connection, you can use the ifconfig command to check the IP address obtained by wlan0.

```

root@ATK-DLRV1126B:/# ifconfig
eth0      Link encap:Ethernet  HWaddr 36:AD:63:E2:79:C4
          UP BROADCAST MULTICAST  MTU:1500  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)
          Interrupt:79

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:65536  Metric:1
          RX packets:2874 errors:0 dropped:0 overruns:0 frame:0
          TX packets:2874 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:175191 (171.0 KiB)  TX bytes:175191 (171.0 KiB)

wlan0     Link encap:Ethernet  HWaddr 4C:A3:8F:7B:40:BA
          inet addr:192.168.7.217 Bcast:192.168.7.255 Mask:255.255.255.0
          inet6 addr: fe80::4ea3:8fff:fe7b:40ba/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:552 errors:0 dropped:138 overruns:0 frame:0
          TX packets:83 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:78031 (76.2 KiB)  TX bytes:10129 (9.8 KiB)

root@ATK-DLRV1126B:/#

```

Figure 2.17-5 wlan0 has obtained an IP address.

Test the connectivity by pinging Baidu, or you can also ping the gateway to test the WiFi connectivity.

```

ping -I wlan0 www.baidu.com # Ping Baidu for testing
Ping -I wlan0 192.168.7.1
# Ping the gateway. Please change it to your own gateway: 192.168.x.x

```

```

root@ATK-DLRV1126B:/# ping -I wlan0 www.baidu.com
PING www.baidu.com (183.2.172.177) from 192.168.7.217 wlan0: 56(84) bytes of data.
64 bytes from 183.2.172.177: icmp_seq=1 ttl=51 time=5.78 ms
64 bytes from 183.2.172.177: icmp_seq=2 ttl=51 time=4.91 ms
64 bytes from 183.2.172.177: icmp_seq=3 ttl=51 time=4.47 ms
64 bytes from 183.2.172.177: icmp_seq=4 ttl=51 time=4.48 ms
64 bytes from 183.2.172.177: icmp_seq=5 ttl=51 time=5.22 ms
64 bytes from 183.2.172.177: icmp_seq=6 ttl=51 time=4.73 ms
64 bytes from 183.2.172.177: icmp_seq=7 ttl=51 time=4.48 ms
64 bytes from 183.2.172.177: icmp_seq=8 ttl=51 time=5.11 ms
64 bytes from 183.2.172.177: icmp_seq=9 ttl=51 time=4.36 ms
64 bytes from 183.2.172.177: icmp_seq=10 ttl=51 time=6.50 ms
^C
--- www.baidu.com ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9013ms
rtt min/avg/max/mdev = 4.364/5.002/6.496/0.649 ms
root@ATK-DLRV1126B:/# ping -I wlan0 192.168.7.1
PING 192.168.7.1 (192.168.7.1) from 192.168.7.217 wlan0: 56(84) bytes of data.
64 bytes from 192.168.7.1: icmp_seq=1 ttl=64 time=2.03 ms
64 bytes from 192.168.7.1: icmp_seq=2 ttl=64 time=1.70 ms
64 bytes from 192.168.7.1: icmp_seq=3 ttl=64 time=1.22 ms
64 bytes from 192.168.7.1: icmp_seq=4 ttl=64 time=1.08 ms
64 bytes from 192.168.7.1: icmp_seq=5 ttl=64 time=1.48 ms
64 bytes from 192.168.7.1: icmp_seq=6 ttl=64 time=1.47 ms
64 bytes from 192.168.7.1: icmp_seq=7 ttl=64 time=1.46 ms
64 bytes from 192.168.7.1: icmp_seq=8 ttl=64 time=1.22 ms
64 bytes from 192.168.7.1: icmp_seq=9 ttl=64 time=1.72 ms
64 bytes from 192.168.7.1: icmp_seq=10 ttl=64 time=1.60 ms
64 bytes from 192.168.7.1: icmp_seq=11 ttl=64 time=1.24 ms
^C
--- 192.168.7.1 ping statistics ---
11 packets transmitted, 11 received, 0% packet loss, time 10016ms
rtt min/avg/max/mdev = 1.083/1.474/2.032/0.265 ms
root@ATK-DLRV1126B:/#

```

Figure 2.17-6 Network Test

2.17.2 SoftAP Mode

SoftAP mode is a working mode of WiFi that allows a WiFi device to function like a wireless router. In SoftAP mode, the WiFi device can act as a hotspot, enabling other devices to connect and use the network services it provides. Generally, SoftAP mode is suitable for scenarios where a device needs to be converted into a wireless hotspot, such as smart home devices, wireless printers, game consoles, etc.

In SoftAP mode, the device can set parameters including SSID and password, allowing other devices to connect and perform data transmission and communication through the network services it provides. Since SoftAP mode enables a device to operate like a wireless router, it is often used in special scenarios such as network debugging and data collection.

The hotspot-enabling scripts developed by ALIENTEK are all stored in the /root/shell/wifi directory, as shown in the figure below.

```

root@ATK-DLRV1126B:/# cd root/shell/wifi/
root@ATK-DLRV1126B:/root/shell/wifi# ls
autoconnect.sh  dnsmasq.conf  hostapd.conf  wifi_apmode.sh
root@ATK-DLRV1126B:/root/shell/wifi#

```

Figure 2.17-7 WiFi Script

The contents of the dnsmasq.conf file are as follows:

```
root@ATK-DLRV1126B:/root/shell/wifi# cat dnsmasq.conf
interface=wlan1
dhcp-range=192.168.4.2,192.168.4.254,255.255.255.0,24h
port=49153
dhcp-leasefile=/userdata/dnsmasq.leases
```

Figure 2.17-8 dnsmasq script

The contents of the hostapd.conf file are as follows:

```
root@ATK-DLRV1126B:/root/shell/wifi# cat hostapd.conf
interface=wlan1
ssid=alientek_softap
driver=nl80211
channel=6
hw_mode=g
ignore_broadcast_ssid=0
auth_algs=1
wpa=3
wpa_passphrase=12345678
wpa_key_mgmt=WPA-PSK
wpa_pairwise=TKIP
rsn_pairwise=CCMP
```

Figure 2.17-9 hostapd configuration file

The contents of the wifi_apmode.sh file are as follows:

```
root@ATK-DLRV1126B:/root/shell/wifi# cat wifi_apmode.sh
if [[ "$(pidof dnsmasq)" != "" ]]
then
    kill -9 $(pidof dnsmasq)
fi

if [[ "$(pidof hostapd)" != "" ]]
then
    kill -9 $(pidof hostapd)
fi

rfkill unblock all
ifconfig wlan1 down
sleep 1
ifconfig wlan1 up
sleep 1
dnsmasq -C dnsmasq.conf
ifconfig wlan1 192.168.4.1
hostapd hostapd.conf -B
```

Figure 2.17-10 wifi_apmode script

Execute the following command to turn on the hotspot. The hotspot is turned on successfully as shown in the figure below.

`./wifi_apmode.sh` # Turn on the hotspot using the script.

```
root@ATK-DLRV1126B:/root/shell/wifi# ./wifi_apmode.sh
[ 3946.771486] IPv6: ADDRCONF(NETDEV_CHANGE): wlan1: link becomes ready
wlan1: interface state UNINITIALIZED->ENABLED
wlan1: AP-ENABLED
root@ATK-DLRV1126B:/root/shell/wifi#
```

Figure 2.17-11 Enable SoftAP mode.

Execute the following command to check whether SoftAP mode has been started.

iw dev # Check whether the type of the wlan device is AP.

```
root@ATK-DLRV1126B:/root/shell/wifi# iw dev
phy#0
    Interface wlan1
        ifindex 6
        wdev 0x2
        addr 4e:a3:8f:7b:40:ba
        ssid alientek_softap
        type AP
        channel 48 (5240 MHz), width: 40 MHz, center1: 5230 MHz
        txpower 19.00 dBm
    Interface wlan0
        ifindex 5
        wdev 0x1
        addr 4c:a3:8f:7b:40:ba
        ssid ALIENTEK-HYS
        type managed
        channel 48 (5240 MHz), width: 40 MHz, center1: 5230 MHz
        txpower 19.00 dBm
root@ATK-DLRV1126B:/root/shell/wifi#
```

Figure 2.17-12 wlan Type

Use your phone to connect and test the hotspot named "alientek_softap". The password is "12345678", which is configured in the hostapd.conf file. I have connected my phone to the hotspot emitted by the board. As you can see, the board's IP is the wlan1 IP 192.168.6.1 set in the wifi-apmode.sh script, acting as the gateway (i.e., the router). The phone has also been assigned an IP address by the development board.

Note that this hotspot can be connected, but it is **not for Internet access**. This demo is only for local area network (LAN) connection.

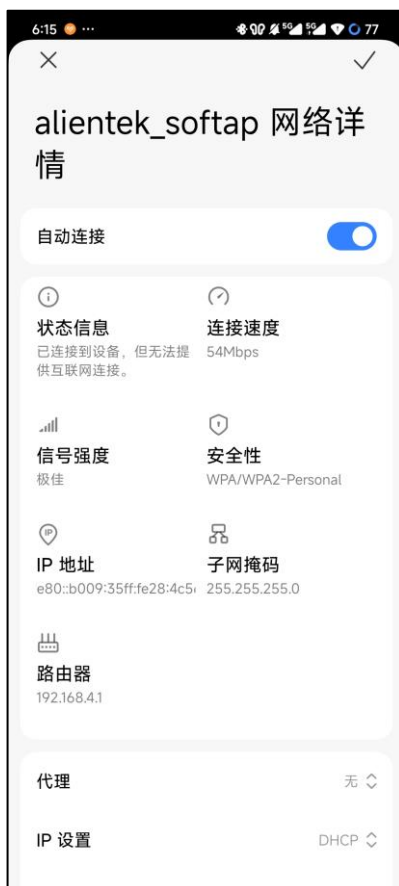


Figure 2.17-13 Mobile phone connects to the development board hotspot.

Now that we have enabled AP mode, how do we disable it? In the previous script, we used hostapd and dnsmasq to start AP mode as background processes, so we need to stop these two processes to disable AP mode. Execute the following commands to stop the processes.

```
killall hostapd dnsmasq
```

At this point, we use iw dev to check the wireless network interface. If the type of the wireless network is **managed**, it means it is in client mode, which is the **station mode** we mentioned above.

```
root@ATK-DLRV1126B:/root/shell/wifi# iw dev
phy#0
    Interface wlan1
        ifindex 6
        wdev 0x2
        addr 4e:a3:8f:7b:40:ba
        type managed
        channel 48 (5240 MHz), width: 40 MHz, center1: 5230 MHz
        txpower -100.00 dBm
    Interface wlan0
        ifindex 5
        wdev 0x1
        addr 4c:a3:8f:7b:40:ba
        ssid ALIENTEK-HYS
        type managed
        channel 48 (5240 MHz), width: 40 MHz, center1: 5230 MHz
        txpower 19.00 dBm
root@ATK-DLRV1126B:/root/shell/wifi#
```

Figure 2.17-14 Normal WiFi Mode

2.18 Onboard USB Bluetooth Test

Bluetooth development relies on Buildroot. Files such as the Bluetooth driver module rtk_btusb.ko are only generated when the correct WiFi/BT module is selected in the Buildroot configuration.

There are many Bluetooth protocols, as listed below:

- **L2CAP**: Logical Link Control and Adaptation Protocol, used to establish and maintain logical link connections.
- **RFCOMM**: Radio Frequency Communication, which maps data transmission of serial port devices to Bluetooth.
- **SDP**: Service Discovery Protocol, used to discover and describe available Bluetooth services.
- **GAP**: Generic Access Profile, defining basic connection and communication rules between devices.
- **GATT**: Generic Attribute Profile, used to establish and manage attribute data interaction between devices.
- **ATT**: Attribute Protocol, used to transmit attribute values between Bluetooth devices.
- **HFP**: Hands-Free Profile, used for audio communication between mobile phones and Bluetooth headsets.
- **A2DP**: Advanced Audio Distribution Profile, used to transmit high-quality audio data between Bluetooth devices.
- **AVRCP**: Audio/Video Remote Control Profile, used to transmit media control commands between Bluetooth devices.

Due to my limited knowledge, Bluetooth is a vast topic. This section only demonstrates the Bluetooth ping test; you may explore other protocols on your own.

The onboard Bluetooth driver module is located in the Buildroot system path:

`/usr/lib/modules/6.1.141/kernel/drivers/bluetooth/bluetooth_usb_driver/rtk_btu
sb.ko`

2.18.1 Check Bluetooth

Enter the following commands:

```
hciconfig
hciconfig hci0 up
Bluetoothctl
[bluetooth]# show
```

```
root@ATK-DLRV1126B:/# hciconfig
hci0: Type: Primary Bus: USB
      BD Address: 4C:A3:8F:7B:C0:BB ACL MTU: 1021:6 SCO MTU: 255:12
      UP RUNNING
      RX bytes:866 acl:0 sco:0 events:70 errors:0
      TX bytes:3283 acl:0 sco:0 commands:70 errors:0

root@ATK-DLRV1126B:/# hciconfig hci0 up
root@ATK-DLRV1126B:/# bluetoothctl
Waiting to connect to bluetoothd...[bluetooth]# hci0 new_settings: powered bondable ssp br/edr le secure-conn wide-band-speech cis-central c
[bluetooth]# Agent registered
[bluetooth]# [CHG] Controller 4C:A3:8F:7B:C0:BB Pairable: yes
[bluetooth]# show
Controller 4C:A3:8F:7B:C0:BB (public)
  Manufacturer: 0x005d (93)
  Version: 0x0b (11)
  Name: BlueZ 5.77
  Alias: BlueZ 5.77
  Class: 0x002c0414 (2884628)
  Powered: yes
  PowerState: on
  Discoverable: no
  DiscoverableTimeout: 0x000000b4 (180)
  Pairable: yes
  UUID: A/V Remote Control (0000110e-0000-1000-8000-00805f9b34fb)
  UUID: Audio Source (0000110a-0000-1000-8000-00805f9b34fb)
  UUID: PnP Information (00001200-0000-1000-8000-00805f9b34fb)
  UUID: Audio Sink (0000110b-0000-1000-8000-00805f9b34fb)
  UUID: Headset (00001108-0000-1000-8000-00805f9b34fb)
  UUID: A/V Remote Control Target (0000110c-0000-1000-8000-00805f9b34fb)
  UUID: Generic Access Profile (00001800-0000-1000-8000-00805f9b34fb)
  UUID: Generic Attribute Profile (00001801-0000-1000-8000-00805f9b34fb)
  UUID: Device Information (0000180a-0000-1000-8000-00805f9b34fb)
  UUID: Headset AG (00001112-0000-1000-8000-00805f9b34fb)
  Modelias: usb:v1D6Bp0246d054D
  Discovering: no
  Roles: central
  Roles: peripheral
Advertising Features:
  ActiveInstances: 0x00 (0)
  SupportedInstances: 0x04 (4)
  SupportedIncludes: tx-power
  SupportedIncludes: appearance
  SupportedIncludes: local-name
  SupportedSecondaryChannels: 1M
  SupportedSecondaryChannels: 2M
  SupportedSecondaryChannels: Coded
  SupportedCapabilities.MinTxPower: 0xfffffff (-17)
  SupportedCapabilities.MaxTxPower: 0x000b (11)
  SupportedCapabilities.MaxAdvLen: 0xfb (251)
  SupportedCapabilities.MaxScnRspLen: 0xfb (251)
  SupportedFeatures: CanSetTxPower
  SupportedFeatures: HardwareOffload
[bluetooth]#
```

Figure 2.18-1 Check Bluetooth device information

Execute exit to quit.

2.18.2 Bluetooth Ping Test

```
hciconfig hci0 piscan
Hcitol scan
```

```

root@ATK-DLRV1126B:/# hciconfig hci0 piscan
root@ATK-DLRV1126B:/# hcitool scan
Scanning ...
[ 182.668090] rtk_btcoex: hci (periodic)inq start
[ 192.917961] rtk_btcoex: inquiry complete
    00:13:EF:A0:04:F1      n/a
    E4:40:97:4A:8B:9C      一加Ace2
    00:A7:50:40:92:92      n/a
    F2:C6:85:76:B2:AA      n/a. 一加Ace2
    F1:55:DB:57:F5:A3      n/a
    48:95:07:53:7F:7E      OPP0 A5
    5D:52:96:85:D9:CF      BLE03_SPP
    44:BF:5C:5B:97:82      n/a
    04:10:6B:D5:41:0F      [REDACTED]
    E4:FD:D8:ED:31:49      BLE03_SPP.
root@ATK-DLRV1126B:/#

```

Figure 2.18-2 Scan for Bluetooth devices.

Use the following command to ping the Bluetooth device with the address 04:10:6B:D5:41:0F.

```
l2ping 04:10:6B:D5:41:0F
```

The result is shown in the figure below:

```

root@ATK-DLRV1126B:/# l2ping 04:10:6B:D5:41:0F
[ 279.421942] rtk_btcoex: hci create connection, start paging
[ 279.852924] rtk_btcoex: connected, handle 0006, status 0x00
[ 279.852949] rtk_btcoex: Page success
Ping: 04:10:6B:D5:41:0F from 4C:A3:8F:7B:C0:BB (data size 44) ...
44 bytes from 04:10:6B:D5:41:0F id 0 time 8.79ms
44 bytes from 04:10:6B:D5:41:0F id 1 time 18.44ms
44 bytes from 04:10:6B:D5:41:0F id 2 time 17.51ms
44 bytes from 04:10:6B:D5:41:0F id 3 time 27.26ms
44 bytes from 04:10:6B:D5:41:0F id 4 time 33.74ms
44 bytes from 04:10:6B:D5:41:0F id 5 time 19.77ms
44 bytes from 04:10:6B:D5:41:0F id 6 time 18.74ms
44 bytes from 04:10:6B:D5:41:0F id 7 time 43.69ms
44 bytes from 04:10:6B:D5:41:0F id 8 time 30.94ms
44 bytes from 04:10:6B:D5:41:0F id 9 time 27.25ms
44 bytes from 04:10:6B:D5:41:0F id 10 time 26.12ms
^C11 sent, 11 received, 0% loss

```

Figure 2.18-3 l2ping Test

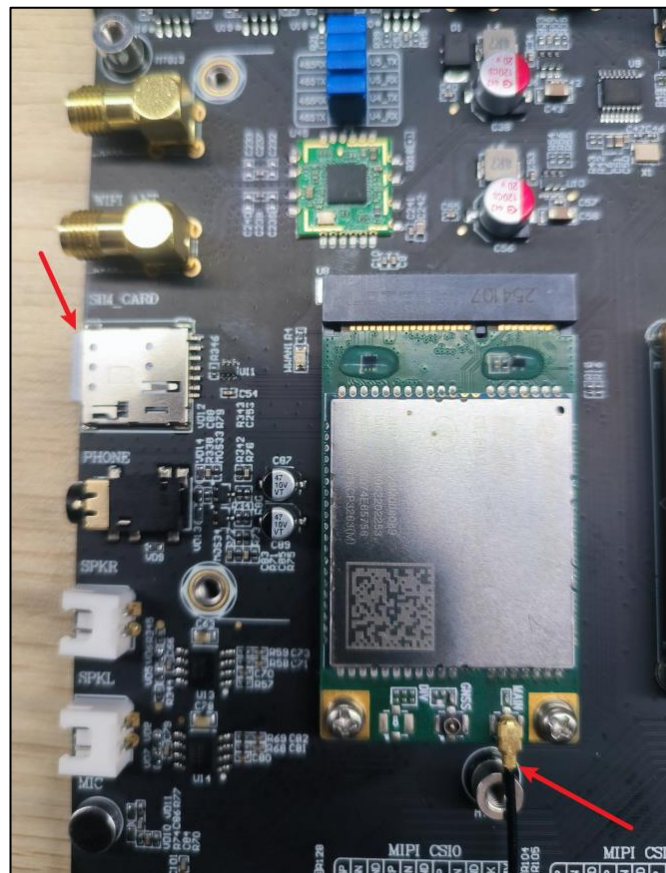
2.19 4G Module Test

The default system of the ATK-DLRV1126B development board is adapted to the Quectel EC20 4G module, as shown in the figure below:



Figure 2.19-14G Module

First, insert the 4G module and SIM card into the development board according to the following picture, and install the module antenna.



Next, we use the following command to check whether the system has detected the module and generated the virtual serial port node:

```
ls /dev/ttyUSB*
```

```
root@ATK-DLRV1126B:/#
root@ATK-DLRV1126B:/# resize
COLUMNS=210;LINES=59;export COLUMNS LINES;
root@ATK-DLRV1126B:/# ls /dev/ttyUSB* -l
crw-rw---- 1 root dialout 188, 0 Jan 1 1970 /dev/ttyUSB0
crw-rw---- 1 root dialout 188, 1 Jan 1 1970 /dev/ttyUSB1
crw-rw---- 1 root dialout 188, 2 Jan 1 1970 /dev/ttyUSB2
crw-rw---- 1 root dialout 188, 3 Jan 1 1970 /dev/ttyUSB3
root@ATK-DLRV1126B:/#
```

Then, use the following command to dial:

```
quctel-CM &
```

```
root@ATK-DLRV1126B:/# quctel-CM &
[1] 795
root@ATK-DLRV1126B:/# [12-15_17:45:54:662] QConnectManager Linux V1.6.8
[12-15_17:45:54:663] Find /sys/bus/usb/devices/3-1.2 idVendor=0x2c7c idProduct=0x125, bus=0x003, dev=0x004
[12-15_17:45:54:663] Auto find qmichannel = /dev/cdc-wdm0
[12-15_17:45:54:663] Auto find usbnet_adapter = wwan0
[12-15_17:45:54:663] netcard driver = qmi_wwan_q, driver version = V1.2.9
[12-15_17:45:54:663] Modem works in QMI mode
[12-15_17:45:54:669] cdc_wdm_fd = 7
[12-15_17:45:54:778] Get clientWDS = 5
[12-15_17:45:54:809] Get clientDMS = 1
[12-15_17:45:54:840] Get clientNAS = 2
[12-15_17:45:54:872] Get clientUIM = 1
[12-15_17:45:54:904] Get clientWDA = 1
[12-15_17:45:54:936] requestBaseBandVersion EC20CEFILGR08A01M1G
[12-15_17:45:55:064] requestGetSIMStatus SIMStatus: SIM_READY
[12-15_17:45:55:129] requestGetProfile[pdp:1 index:1] CTNET///0/IPV4V6
[12-15_17:45:55:160] requestRegistrationState2 MCC: 460, MNC: 11, PS: Attached, DataCap: LTE
[12-15_17:45:55:192] requestQueryDataCall IPv4ConnectionStatus: DISCONNECTED
[12-15_17:45:55:192] ip addr flush dev wwan0
[12-15_17:45:55:200] ip link set dev wwan0 down
[12-15_17:45:55:257] requestSetupDataCall WdsConnectionIPv4Handle: 0x86d142f0
[12-15_17:45:55:385] ip link set dev wwan0 up
[12-15_17:45:55:394] busybox udhcpc -f -n -q -t 5 -i wwan0
udhcpc: started, v1.36.1
udhcpc: broadcasting discover
udhcpc: broadcasting select for 10.10.42.54, server 10.10.42.53
udhcpc: lease of 10.10.42.54 obtained from 10.10.42.53, lease time 7200
[12-15_17:45:55:532] deleting routers
[12-15_17:45:55:548] adding dns 202.96.128.86
[12-15_17:45:55:549] adding dns 202.96.134.133
```

If the dial-up is successful, run the ifconfig command and you will see that the network interface of the 4G module has obtained an IP address, as shown in the figure below:


```

root@ATK-DLRV1126B:/# ifconfig
eth0      Link encap:Ethernet  HWaddr F6:98:74:4F:F6:44
          UP BROADCAST MULTICAST  MTU:1500  Metric:1
          RX packets:0  errors:0  dropped:0  overruns:0  frame:0
          TX packets:0  errors:0  dropped:0  overruns:0  carrier:0
          collisions:0  txqueuelen:1000
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)
          Interrupt:79

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:65536  Metric:1
          RX packets:94  errors:0  dropped:0  overruns:0  frame:0
          TX packets:94  errors:0  dropped:0  overruns:0  carrier:0
          collisions:0  txqueuelen:1000
          RX bytes:9090 (8.8 KiB)  TX bytes:9090 (8.8 KiB)

wlan0     Link encap:Ethernet  HWaddr DC:84:03:CB:23:83
          inet addr:192.168.7.76  Bcast:192.168.7.255  Mask:255.255.255.0
          inet6 addr: fe80::de84:3fff:fe2b:2383/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:615  errors:0  dropped:45  overruns:0  frame:0
          TX packets:73  errors:0  dropped:0  overruns:0  carrier:0
          collisions:0  txqueuelen:1000
          RX bytes:57758 (56.4 KiB)  TX bytes:8547 (8.3 KiB)

wwan0     Link encap:Ethernet  HWaddr 72:E2:67:DD:1E:28
          inet addr:10.10.42.54  Mask:255.255.255.252
          inet6 addr: fe80::70e2:67ff:fedd:1e28/64 Scope:Link
          UP RUNNING NOARP  MTU:1500  Metric:1
          RX packets:2  errors:0  dropped:0  overruns:0  frame:0
          TX packets:7  errors:0  dropped:0  overruns:0  carrier:0
          collisions:0  txqueuelen:1000
          RX bytes:612 (612.0 B)  TX bytes:996 (996.0 B)

root@ATK-DLRV1126B:/#

```

Finally, we can execute the following ping command to test whether the network can be accessed normally:

```
ping www.baidu.com
```

```

root@ATK-DLRV1126B:/#
root@ATK-DLRV1126B:/# ping -I wwan0 www.baidu.com
PING www.a.shifen.com (183.2.172.17) from 10.10.42.54 wwan0: 56(84) bytes of data.
64 bytes from 183.2.172.17: icmp_seq=1 ttl=52 time=121 ms
64 bytes from 183.2.172.17: icmp_seq=2 ttl=52 time=24.5 ms
64 bytes from 183.2.172.17: icmp_seq=3 ttl=52 time=22.7 ms
64 bytes from 183.2.172.17: icmp_seq=4 ttl=52 time=56.9 ms
64 bytes from 183.2.172.17: icmp_seq=5 ttl=52 time=57.1 ms
64 bytes from 183.2.172.17: icmp_seq=6 ttl=52 time=55.4 ms
64 bytes from 183.2.172.17: icmp_seq=7 ttl=52 time=54.1 ms
64 bytes from 183.2.172.17: icmp_seq=8 ttl=52 time=52.8 ms
64 bytes from 183.2.172.17: icmp_seq=9 ttl=52 time=50.8 ms
64 bytes from 183.2.172.17: icmp_seq=10 ttl=52 time=48.8 ms
64 bytes from 183.2.172.17: icmp_seq=11 ttl=52 time=46.5 ms
64 bytes from 183.2.172.17: icmp_seq=12 ttl=52 time=44.6 ms
64 bytes from 183.2.172.17: icmp_seq=13 ttl=52 time=43.3 ms
64 bytes from 183.2.172.17: icmp_seq=14 ttl=52 time=41.9 ms
^C
--- www.a.shifen.com ping statistics ---
14 packets transmitted, 14 received, 0% packet loss, time 13021ms
rtt min/avg/max/mdev = 22.732/51.425/120.648/21.816 ms
root@ATK-DLRV1126B:/#

```

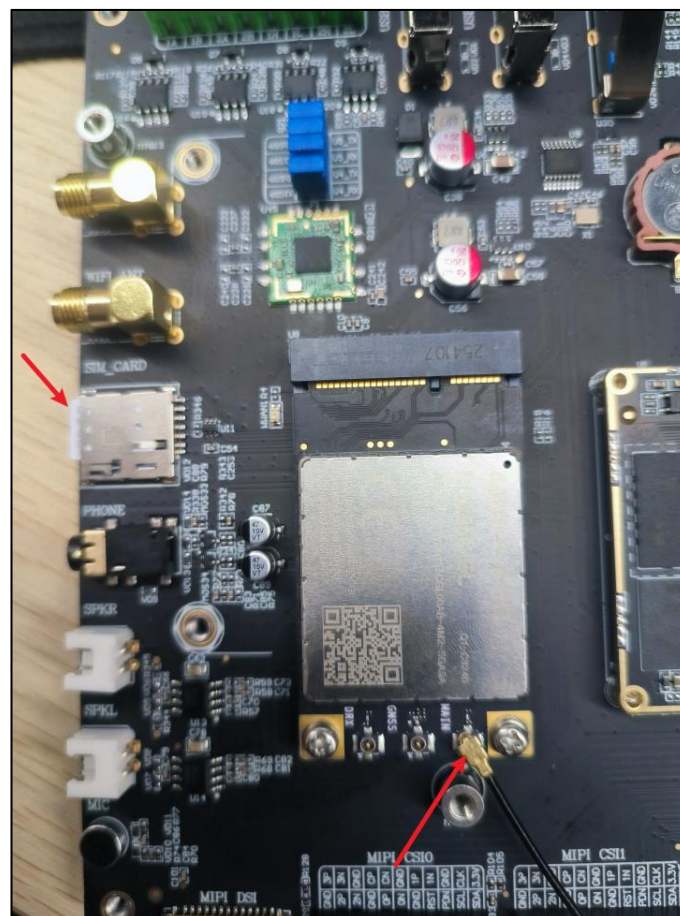
2.20 5G Module Test

The factory system of the ATK-DLRV1126B development board supports the Quectel RG255C 5G module, as shown in the figure below:



Figure 2.20-15G Module

First, insert the 5G module and the SIM card into the development board according to the following picture, and then install the module antenna.



Next, we use the following command to check whether the system has detected the module and generated the virtual serial port nodes:

```
ls /dev/ttyUSB*
```

```
root@ATK-DLRV1126B:/#
root@ATK-DLRV1126B:/# ls /dev/ttyUSB* -l
crw-rw---- 1 root dialout 188, 0 Jan  1 08:00 /dev/ttyUSB0
crw-rw---- 1 root dialout 188, 1 Jan  1 08:00 /dev/ttyUSB1
crw-rw---- 1 root dialout 188, 2 Jan  1 08:00 /dev/ttyUSB2
root@ATK-DLRV1126B:/#
```

Then, use the following command to perform dial-up:

```
quectel-CM &
```

```
root@ATK-DLRV1126B:/# quectel-CM &
[1] 800
root@ATK-DLRV1126B:/# [12-15 17:54:37:877] QConnectManager_Linux_V1.6.8
[12-15 17:54:37:878] Find /sys/bus/usb/devices/3-1.2 idVendor=0x2c7c idProduct=0x316, bus=0x003, dev=0x004
[12-15 17:54:37:879] Auto find qmichannel = /dev/cdc-wdm0
[12-15 17:54:37:879] Auto find usbnet_adapter = wwan0
[12-15 17:54:37:879] netcard driver = qmi_wwan_q, driver version = V1.2.9
[12-15 17:54:37:879] qmap_mode = 1, qmap_version = 5, qmap_size = 16384, muxid = 0x81, qmap_netcard = wwan0
[12-15 17:54:37:879] Modem works in QMI mode
[12-15 17:54:37:885] cdc_wdm_fd = 7
[12-15 17:54:38:885] QmiWwanInit message timeout
[12-15 17:54:40:158] Get clientWDS = 9
[12-15 17:54:40:190] Get clientDMS = 1
[12-15 17:54:40:222] Get clientNAS = 3
[12-15 17:54:40:254] Get clientUIM = 2
[12-15 17:54:40:286] Get clientWDA = 1
[12-15 17:54:40:318] requestBaseBandVersion RG255CGLABR01A04M4G
[12-15 17:54:40:350] qmap_settings.rx_urb_size = 16384
[12-15 17:54:40:350] qmap_settings.ul_data_aggregation_max_datagrams = 11
[12-15 17:54:40:350] qmap_settings.ul_data_aggregation_max_size = 8192
[12-15 17:54:40:350] qmap_settings.dl_minimum_padding = 0
[12-15 17:54:40:478] requestGetSIMStatus SIMStatus: SIM_READY
[12-15 17:54:40:510] requestRegistrationState2 MCC: 460, MNC: 11, PS: Attached, DataCap: 5G_SA
[12-15 17:54:40:542] requestQueryDataCall IPv4ConnectionStatus: DISCONNECTED
[12-15 17:54:40:542] ip addr flush dev wwan0
[12-15 17:54:40:549] ip link set dev wwan0 down
[12-15 17:54:40:926] requestSetupDataCall WdsConnectionIPv4Handle: 0xe08d7ee0
[12-15 17:54:41:016] net wwan0: link state 0x0 -> 0x1
[12-15 17:54:41:054] ip link set dev wwan0 up
[12-15 17:54:41:062] IPv6: ADDRCONF(NETDEV_CHANGE): wwan0: link becomes ready
[12-15 17:54:41:062] ip -4 address flush dev wwan0
[12-15 17:54:41:069] ip -4 address add 10.4.135.187/29 dev wwan0
[12-15 17:54:41:077] ip -4 route add default via 10.4.135.188 dev wwan0
RTNETLINK answers: File exists
```

If the dial-up is successful, run the ifconfig command and you will see that the network interface of the 4G module has obtained an IP address, as shown in the figure below:

```

root@ATK-DLRV1126B:/# ifconfig
eth0      Link encap:Ethernet  HWaddr F6:98:74:4F:F6:44
          UP BROADCAST MULTICAST  MTU:1500  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)
          Interrupt:79

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:65536  Metric:1
          RX packets:102 errors:0 dropped:0 overruns:0 frame:0
          TX packets:102 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:9958 (9.7 KiB)  TX bytes:9958 (9.7 KiB)

wlan0     Link encap:Ethernet  HWaddr DC:84:03:CB:23:83
          inet addr:192.168.7.76  Bcast:192.168.7.255  Mask:255.255.255.0
          inet6 addr: fe80::de84:3ff:fe8b:2383/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:134 errors:0 dropped:37 overruns:0 frame:0
          TX packets:76 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:18735 (18.2 KiB)  TX bytes:8961 (8.7 KiB)

wwan0     Link encap:Ethernet  HWaddr 76:D0:48:8C:49:55
          inet addr:10.4.135.187  Mask:255.255.255.248
          inet6 addr: fe80::74d0:48ff:fe8c:4955/64 Scope:Link
          UP RUNNING NOARP  MTU:1500  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:2 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:0 (0.0 B)  TX bytes:120 (120.0 B)

root@ATK-DLRV1126B:/#

```

Finally, we can execute the following ping command to test whether the network can be accessed normally:

ping www.baidu.com

```

root@ATK-DLRV1126B:/# ping -I wwan0 www.baidu.com
PING www.a.shifen.com (183.2.172.17) from 10.4.135.187 wwan0: 56(84) bytes of data.
64 bytes from 183.2.172.17: icmp_seq=1 ttl=52 time=160 ms
64 bytes from 183.2.172.17: icmp_seq=2 ttl=52 time=33.8 ms
64 bytes from 183.2.172.17: icmp_seq=3 ttl=52 time=31.8 ms
64 bytes from 183.2.172.17: icmp_seq=4 ttl=52 time=29.8 ms
64 bytes from 183.2.172.17: icmp_seq=5 ttl=52 time=27.8 ms
64 bytes from 183.2.172.17: icmp_seq=6 ttl=52 time=25.9 ms
64 bytes from 183.2.172.17: icmp_seq=7 ttl=52 time=24.8 ms
64 bytes from 183.2.172.17: icmp_seq=8 ttl=52 time=22.8 ms
64 bytes from 183.2.172.17: icmp_seq=9 ttl=52 time=40.8 ms
64 bytes from 183.2.172.17: icmp_seq=10 ttl=52 time=38.8 ms
64 bytes from 183.2.172.17: icmp_seq=11 ttl=52 time=36.8 ms
64 bytes from 183.2.172.17: icmp_seq=12 ttl=52 time=34.8 ms
64 bytes from 183.2.172.17: icmp_seq=13 ttl=52 time=32.8 ms
64 bytes from 183.2.172.17: icmp_seq=14 ttl=52 time=30.8 ms
64 bytes from 183.2.172.17: icmp_seq=15 ttl=52 time=28.8 ms
64 bytes from 183.2.172.17: icmp_seq=16 ttl=52 time=26.8 ms
64 bytes from 183.2.172.17: icmp_seq=17 ttl=52 time=24.8 ms
64 bytes from 183.2.172.17: icmp_seq=18 ttl=52 time=22.8 ms
64 bytes from 183.2.172.17: icmp_seq=19 ttl=52 time=40.8 ms
64 bytes from 183.2.172.17: icmp_seq=20 ttl=52 time=38.8 ms
64 bytes from 183.2.172.17: icmp_seq=21 ttl=52 time=277 ms
64 bytes from 183.2.172.17: icmp_seq=22 ttl=52 time=222 ms
64 bytes from 183.2.172.17: icmp_seq=23 ttl=52 time=196 ms
64 bytes from 183.2.172.17: icmp_seq=24 ttl=52 time=30.5 ms
64 bytes from 183.2.172.17: icmp_seq=25 ttl=52 time=32.3 ms
64 bytes from 183.2.172.17: icmp_seq=26 ttl=52 time=31.3 ms
64 bytes from 183.2.172.17: icmp_seq=27 ttl=52 time=29.8 ms
64 bytes from 183.2.172.17: icmp_seq=28 ttl=52 time=27.8 ms
64 bytes from 183.2.172.17: icmp_seq=29 ttl=52 time=25.8 ms
64 bytes from 183.2.172.17: icmp_seq=30 ttl=52 time=23.8 ms
64 bytes from 183.2.172.17: icmp_seq=31 ttl=52 time=41.8 ms
64 bytes from 183.2.172.17: icmp_seq=32 ttl=52 time=39.8 ms
64 bytes from 183.2.172.17: icmp_seq=33 ttl=52 time=37.8 ms
64 bytes from 183.2.172.17: icmp_seq=34 ttl=52 time=35.8 ms
64 bytes from 183.2.172.17: icmp_seq=35 ttl=52 time=38.9 ms
64 bytes from 183.2.172.17: icmp_seq=36 ttl=52 time=32.8 ms
64 bytes from 183.2.172.17: icmp_seq=37 ttl=52 time=30.7 ms
64 bytes from 183.2.172.17: icmp_seq=38 ttl=52 time=28.7 ms

```

